

Master in Intelligent Interactive Systems
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Synthetic Mobility Data: A User-Level Membership Inference Assessment

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Abstract

Wider access to mobility data would advance urban planning and transport modelling; such data is rarely shared because movement records expose sensitive information about individuals. Deep generative models could promise a solution: synthetic datasets that retain analytical utility while protecting the people in the training data. Whether they can deliver both desired attributes together remains unclear, prior work has largely examined tabular data or record-level privacy rather than the user level. This study evaluates the privacy–utility trade-off of a Variational Autoencoder and a Wasserstein GAN against a bootstrap resampling baseline, applied to origin–destination trips derived from the GeoLife GPS dataset (7,739 trips from 154 users in Beijing). The research measures (i) how faithfully the synthetic data reproduces real mobility, (ii) whether a user-level membership inference attack can identify individuals among the training set, and (iii) whether more active users are more exposed. The models reproduced temporal structure well but spatial structure poorly, recovering only around a quarter of real-data performance on a downstream prediction task (TSTR ratio 0.25–0.27). Neither showed measurable membership leakage ($\text{AUC} \approx 0.50$), against a bootstrap positive control that leaked clearly ($\text{AUC} 0.75$). No membership-based difference across activity levels could be established, though direct record reproduction rose with user activity across all models. These results indicate that the observed privacy is largely a by-product of under-fitting rather than evidence of a favourable trade-off: the implemented models preserved privacy at the cost of utility. Rigorously establishing where privacy and utility can coexist is a necessary foundation for releasing synthetic mobility data with confidence, increasing open access that mobility research depends on.

Keywords: Mobility Origin-Destination Trips, Synthetic Data Generation, Spatio-temporal Data, Privacy–Utility Evaluation, Membership Inference Attacks

Chapter 1

Introduction

1.1 Motivation

The Value of Data

Data enables us to understand complex systems, identify patterns that would otherwise remain hidden, and make evidence-based decisions. Throughout history, access to data has enabled major advances in science and public policy. During the Crimean War in 1854, Florence Nightingale systematically collected mortality records and demonstrated that far more British soldiers were dying from preventable diseases such as cholera, typhus and dysentery than from battlefield wounds. By presenting these data through innovative visualisations, she was able to challenge prevailing assumptions and advocate for major reforms in military healthcare. (Bradshaw 2020)

Similarly, around the same time during the 1854 cholera outbreak in London, John Snow collected and analysed data on cholera deaths, linking cases to specific water sources. Snow demonstrated that contaminated drinking water, rather than poor air quality, was driving the epidemic. His analysis identified the Broad Street pump as a key source of infection and provided evidence that influenced improvements to sanitation and water infrastructure. (Tulchinsky 2018)

The importance of access to data extends beyond healthcare. One of the most obvious examples is environmental monitoring, the notable case being the identification and investigation of climate change, shown to be causing a 0.9 °C rise in global sea surface temperatures. (ECMWF n.d.)

More recent transformative examples can be found in professional sports, where the teams have used statistical analysis to identify undervalued players and compete effectively against significantly wealthier teams (Shughart 2004). Data-driven strategies are now common throughout sports, in areas where statistical analysis was previously considered irrelevant.

These examples illustrate a broader principle: meaningful access to data enables the understanding of real-world systems, and crucially enables the delivery of intelligent solutions, advantages or benefits.

The privacy problem

Despite the importance of data-driven research, access to high-quality datasets remains limited across many domains. Privacy concerns, legal restrictions, commercial sensitivity, and ethical considerations often prevent organisations from releasing datasets publicly. Mobility data consists of spatio-temporal records describing how people move through space and time, typically collected through GPS devices, mobile phones, or transport systems. Such data is critical for applications including urban planning, traffic forecasting, and epidemic modelling. Prior work has shown that human mobility is not purely random, but instead exhibits strong temporal and spatial regularity, with individuals frequently returning to a small set of highly visited locations (González, Hidalgo, and Barabási 2008)

However, mobility datasets are rarely openly available because movement patterns can reveal sensitive information about individuals, including home locations, workplaces, routines, and social behaviour.

Synthetic Data as a Solution

Synthetic data generation has emerged as a promising approach for addressing this challenge. Rather than releasing original records, generative models are trained on real datasets and used to produce artificial data that preserves important statistical patterns while avoiding direct disclosure of individual information. Recent advances in deep generative modelling, including Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), have enabled increasingly realistic synthetic datasets across a range of domains. However, realistic synthetic data does not necessarily imply privacy protection. A growing body of research has shown that generative models may memorise aspects of their training data, potentially exposing sensitive information through attacks such as membership inference. While synthetic data generation offers a potential solution to the data-access problem, the extent to which generative models successfully balance utility and privacy remains unclear. Existing studies have shown that synthetic datasets can retain valuable information for downstream tasks, yet they may also leak information about the individuals whose data was used during training. Although differential privacy offers strong theoretical privacy guarantees, implementing these guarantees in deep learning models often requires difficult trade-offs between privacy, utility, and model performance. Consequently, many synthetic data generation approaches operate without formal privacy guarantees (Stadler, Oprisanu, and Troncoso 2022). Furthermore, relatively little work has examined this trade-off for origin-destination mobility data at the user level.

Understanding the privacy-utility trade-off in synthetic mobility data is important for both researchers and practitioners. If synthetic data can provide meaningful privacy protection while preserving analytical value, organisations may be able to share datasets more openly, accelerating research and innovation. Conversely, if generative models leak information about individuals, synthetic data releases may create a false sense of privacy. Evaluating these risks is therefore essential before synthetic mobility datasets can be considered a reliable mechanism for open data sharing.

1.2 Objectives & Contributions

This research investigates whether deep generative models can produce synthetic origin–destination mobility data that is both useful and private. The study is structured around three research questions:

- RQ1** Can generative models accurately reproduce origin–destination mobility patterns from real trip data?
- RQ2** To what extent do these models leak information about individual users through membership inference attacks?
- RQ3** Are users with higher activity at greater risk of privacy leakage than less active users?

The first question concerns utility and how closely the generated synthetic data preserves the spatial and temporal structure of real mobility. The second and third concern privacy: whether membership can be inferred at all, and whether that risk falls unevenly across users. Together they allow the privacy and utility of each model to be assessed side by side, rather than in isolation. In addressing these questions, this research makes the following contributions:

- A direct comparison of two deep generative architectures (VAE & WGAN-GP) against a Bootstrap resampling baseline.
- A user-level membership inference evaluation using a shadow-model attack with a copying baseline as a positive control, establishing that the attack and representation can detect leakage when it is present.
- An analysis of how privacy risk varies with user activity level, testing directly whether the most active users are the most exposed.
- An evaluation conducted on sparse origin–destination data with limited per-user records.

- Use of an openly available dataset (GeoLife), supporting reproducibility in a field where comparable studies can rely on private data.

1.3 Structure of the Report

The remainder of this report is organised as follows. Chapter 2 reviews the relevant background, covering how mobility data can be represented, the main deep learning architectures that are used to generate synthetic data, the privacy risks posed, as well as any prior work on the privacy–utility trade-off in generative synthesis. Chapter 3 sets out the methodology: the GeoLife dataset used, the preprocessing steps taken to obtain origin–destination trips, the two generative models implemented (VAE & WGAN-GP), and the evaluation framework used to assess utility and perform the user-level membership-inference attack. Chapter 4 presents the results, addressing each research question in turn—reproduction of real mobility patterns (RQ1), membership leakage under attack (RQ2), and whether privacy risk varies with user activity level (RQ3). Chapter 5 interprets these findings, as well as discussing any limitations of the study. Supporting material is provided in the appendices, including further detail on data pre-processing, model training, as well as additional synthetic data spatial plots.

Chapter 2

Background

2.1 Mobility Representation

Representation of mobility data is a critical aspect of how data is encoded but also what information deep learning models learn.

Mobility data can represent different entities and abstractions, from trip trajectories and user GPS trajectories, to user entry and exit network timestamps such as with a metro network. In these cases, the entity that the data represents is a single user or vehicle, however data can be aggregated together to represent a collection of users or vehicles, such as network flows between metro stations. These different representations encode different levels of information, which directly affects both model performance and privacy risk, Table 1 shows the level representation summary.

Data can be converted to higher-level of abstraction (e.g. GPS trajectories to OD trips) but cannot be converted to lower-abstraction levels. The data capture process naturally determines the most granular version of representation. For example, metro tolls could record user entry and exit points on a network, it cannot be converted to more granular trajectory information that encodes user-related movement behaviour, but could however be converted to aggregated flows such as metro stop hourly network flows.

Table 1: Granularity levels of mobility data representation.

Level	Representation	Meaning
Macro	Aggregated flows	Population-level mobility trends
Meso	OD trips	Individual travel patterns (movement between two locations)
Micro	GPS trajectories	Exact individual movement paths

The entity recording the mobility often determines the representation level, transport networks will naturally provide information in the form of user OD trips or aggregated flows. In contrast GPS trajectories normally require active user participation or volunteers, but not always.

While raw GPS trajectories provide rich behavioural information, they introduce significant modelling complexity and privacy risks. Additionally, the granularity of movement trajectories can be more susceptible to noise, as GPS logger accuracy can vary significantly in dense urban environments, and can include dropouts in signal and trajectory information. While some downstream application may require trajectory information, many do not. In contrast, aggregated representations improve privacy but limit their usefulness for certain tasks.

Origin–destination representations offer a middle ground, preserving key mobility structures while remaining tractable for modelling. However, their implications for user-level privacy leakage in generative models remain insufficiently explored, motivating the focus of this study.

In summary, mobility data representation can be viewed as a trade-off between fidelity and abstraction, where lower-level representations preserve detailed behavioural signals while higher-level representations prioritise scalability and privacy.

2.2 Generative Architectures

Different unsupervised architectures have been used to generate synthetic data with different challenges for each. This section looks at each type and summarises the related research. The key feature for all of them is the ability to learn patterns in

training data and to be able to generate synthetic data which is representative of the original pattern. The current architectures can be grouped as the following: Latent Variable Models, Adversarial Models, Hybrid Models, and Diffusion / Score-Based Models

Latent Variable Models The Variational Autoencoder (VAE), introduced by Kingma and Welling (Kingma and Welling 2022), is a latent-variable generative model that learns to encode data into a continuous latent space and decode samples from it back into the data domain. It pairs a probabilistic encoder, which approximates the posterior over latent variables given an input, with a decoder that reconstructs the input from a latent code.

The model is trained by using a reconstruction term that rewards accurate decoding, as well as a Kullback–Leibler (KL) divergence term that regularises the approximate posterior towards a Gaussian prior.

Adversarial Models Generative Adversarial Networks (GANs) take an adversarial rather than a likelihood-based approach to generation. Two networks are trained in competition: a generator maps random noise to synthetic samples, while a discriminator learns to distinguish synthetic samples from real ones. The generator is trained to fool the discriminator, and at convergence produces samples that the discriminator can no longer separate from real data. This adversarial framing tends to yield sharper, more realistic outputs than latent-variable models, but it is also notoriously unstable to train, suffering from problems such as vanishing gradients and mode collapse, in which the generator captures only a narrow part of the data distribution. (Goodfellow et al. 2014)

The Wasserstein GAN (WGAN) was introduced to address this instability. Rather than training the discriminator as a classifier, WGAN trains a critic to approximate the Wasserstein (Earth-Mover) distance between the real and synthetic distributions — informally, the minimum cost of transporting probability mass to transform one distribution into the other. This distance provides smoother, more informative gradients to the generator even when the two distributions barely overlap, and the

critic’s output correlates more reliably with sample quality than the discriminator score of a standard GAN. The formulation requires the critic to be 1-Lipschitz, a constraint the original WGAN enforces by clipping the critic’s weights to a fixed range. (Arjovsky, Chintala, and Bottou 2017)

This weight clipping was found to introduce its own difficulties, biasing the critic towards overly simple functions and causing gradients to vanish or explode. WGAN with gradient penalty (WGAN-GP) (Gulrajani et al. 2017) replaces clipping with a softer constraint: a penalty term that encourages the norm of the critic’s gradient to stay close to one, evaluated at points interpolated between real and synthetic samples. This yields markedly more stable training across a wide range of architectures with little hyperparameter tuning, and is the adversarial model adopted in this study. While these adversarial methods avoid the over-smoothing characteristic of latent-variable models, they provide no explicit probability density and remain more sensitive to training dynamics — a trade-off in modelling behaviour directly relevant to the spatial fidelity results reported later in this work.

Adversarial methods have also been specialised for sequential and temporal data. TimeGAN (Jinsung Yoon, Daniel Jarrett, and Mihaela van der Schaar n.d.), for example, augments the generator–discriminator framework with an autoencoding embedding and a stepwise supervised loss that encourages the model to reproduce temporal dynamics rather than only the marginal appearance of sequences, and has been shown to preserve utility for downstream forecasting and classification tasks. Such approaches are most relevant where fine-grained temporal trajectories must be modelled; in this work, where mobility is represented as origin–destination trips rather than full trajectories, the simpler feed-forward generator described above was sufficient.

Hybrid Models A hybrid VAE-GAN approach has been applied in the context of smart home energy management to generate synthetic datasets for downstream decision-making tasks (Razghandi et al. 2024). The study demonstrates that VAE-GAN outperforms both vanilla GAN and Gaussian Mixture Model (GMM) baselines,

producing synthetic data that enables Q-learning models to achieve performance comparable to that obtained using real data. This suggests that hybrid architectures can effectively capture complex data distributions while preserving utility. However, the study does not evaluate the extent to which the generated data mitigates privacy risks, leaving open questions regarding potential information leakage.

Diffusion / Score-Based Models Diffusion models are a more recent class of generative model that has produced state-of-the-art results, particularly for image synthesis (Ho, Jain, and Abbeel 2020). Rather than mapping noise to data in a single step, they define a forward process that gradually corrupts training samples with Gaussian noise and train a network to reverse it, denoising iteratively until a sample is recovered. This formulation avoids both the adversarial instability of GANs and the over-smoothing characteristic of latent-variable models, and is well suited to capturing the sharp, multimodal distributions that arise in spatial data. Diffusion models were not evaluated here as this study deliberately compares a latent-variable model with an adversarial model, to characterise the privacy–utility trade-off across contrasting training objectives before extending to newer architectures. Given the difficulty both models had in reproducing spatial structure, however, diffusion-based generation is a promising direction for future work.

2.3 Privacy Risks & Attacks

Data privacy, in this context, concerns whether information about the individuals in a training set can be recovered from a model built on their data. Models can leak such information because they do not only learn general patterns, but they can also memorise specific training examples, and this memorisation is what an attacker exploits. Salem et al. (2023) provide a unified treatment of these attacks as a privacy game between a model and an adversary, distinguishing several disclosure risks: membership inference – where the adversary determines whether a particular record or user was part of the training data; attribute inference – where unknown attributes of a record are reconstructed from partial knowledge; and property and reconstruction attacks – which target the dataset as a whole.

Membership inference is among one of the most studied of these risks and serves as a standard proxy for privacy leakage, since a model that reveals who was in its training data has already failed to protect them. Using the synthetic generated dataset, it asks the simple question: was this specific person’s data used to build the dataset?

Attacks are further described by how much information access the adversary has: a white-box attacker can inspect the model’s internals, whereas a black-box attacker sees only its outputs. For this specific research, the focus is on black-box attacks - only access to the model’s outputs (synthetic dataset) and not any of the internals (architecture/model weights).

How successful such an attack can be depends heavily on how much data each individual contributes. An attacker needs enough signal to distinguish one person from another, in the same way that identifying an author from their writing style requires many pages rather than a single sentence. This also shapes the level at which an attack is run. A trip-level attack asks whether a single record was in the training data, while a user-level attack pools all of a person’s records and asks whether if that person was present. Because users who travel more contribute more records, they offer the attacker more to work with, which raises the question whether the most active users are also the most exposed (corresponding to RQ3).

Ideally, privacy-preserving model generalises: it learns and reproduces high-level patterns without copying the exact training samples that would expose individuals. Differential privacy carefully injects noise during training, so that any single individual’s contribution is indistinguishable from within the crowd, providing mathematically bounded privacy guarantees (Dwork 2006). While the technique is regarded as the gold standard in certain applications, such as census publications, Abadi et al. (2016) describe how its application to deep learning is difficult in practice, requiring strict assumptions and careful tuning that often comes at a heavy cost to utility. Differential Privacy is considered out of scope for this research, but its potential to improve privacy and provide guarantees should be stated.

Other defences act at different points in the pipeline. Some intervene during training, by limiting overfitting through regularisation and early stopping. Others intervene after generation, applying statistical disclosure control (SDC) mechanisms such as adding randomness to a model’s outputs or via post-hoc filtering, whereby synthetic records identified as suspiciously close to a real training record — for instance via an Euclidean distance threshold — are perturbed or discarded before release. Filtering of this kind, however, only guards against the specific similarity metric it is built on, and offers no guarantee against leakage through other channels. Underlying all of these is a recurring tension – measures that protect privacy tend to reduce the usefulness of the data, a trade-off that the literature has documented, as described in the next chapter.

2.4 Prior Work: Privacy-Preserving Generative Synthesis

This section reviews prior work studying the privacy–utility trade-off of generative models, moving from those conducted on tabular data towards synthetic mobility data, those closest to this research.

The study *Privacy Preserving Synthetic Data Release Using Deep Learning* (Abay et al. 2019) represents an early exploration of deep generative models for privacy-preserving data synthesis. The authors introduce the DP-EM model, demonstrating that it can achieve improved privacy protection while maintaining competitive data utility compared to traditional approaches. This work also highlights the broader landscape of differentially private methods, including PrivateSVM, PrivateBayes, and DP-VAE, and emphasises the role of differential privacy as a mechanism for providing formal guarantees by limiting the influence of individual data points on model outputs. However, the evaluation is conducted primarily on tabular datasets, and these approaches may not generalise effectively to more complex spatial or temporal data, leaving open questions regarding their applicability to mobility data.

The study *Anonymisation Groundhog Day* (Stadler, Oprisanu, and Troncoso 2022)

provides a comprehensive review of the privacy–utility trade-off in machine learning-based data anonymisation. The authors find that many machine learning approaches fail to provide satisfactory privacy protection while simultaneously degrading data utility, highlighting a fundamental tension between these objectives. Crucially, they also show that privacy protection is not evenly distributed across individuals: outliers receive significantly less benefit, meaning certain users remain more vulnerable even after anonymisation. This observation of uneven, per-individual risk directly motivates this study’s focus on whether more active users are more exposed (RQ3). The authors further note the unpredictability of machine learning models in this context and the lack of formal guarantees compared to differential privacy. As with the previous work, however, the analysis is limited to tabular data, leaving open how these trade-offs manifest in spatiotemporal datasets such as mobility data.

The study *This Person (Probably) Exists* (Webster et al. 2021) examines the vulnerability of generative models to membership inference in a black-box setting, attempting to determine whether specific individuals were part of a GAN’s training data from its outputs alone. The authors show that such leakage is achievable without access to model internals, and, most relevant here, that it is uneven across individuals: a lack of identity diversity and an imbalance in the number of samples per identity both raise the chance of a successful attack, with individuals contributing more data being consistently easier to identify. Although demonstrated on image data, this provides a concrete precedent for the differential per-individual risk that RQ3 investigates in the mobility setting. The authors further find that mitigations such as early stopping reduce leakage by limiting overfitting but do not eliminate it.

WorldMove (Yuan et al. 2026) represents one of the most ambitious efforts in privacy-preserving synthetic mobility data generation, modelling movement patterns across thousands of cities using large-scale aggregated data. The authors, who used a diffusion-based model, demonstrate that the generated data maintains a high level of realism while preserving aggregate mobility patterns. Importantly, their evaluation shows that membership inference attacks are strongly mitigated, with attack performance remaining below 0.6, suggesting a good degree of privacy protection.

This low-leakage result is notable because it echoes the present study’s findings, though it is achieved by very different means: WorldMove relies on large-scale aggregation across many cities, rather than on the behaviour of a model fitted to individual trips. Its limitations include not accounting for seasonal variation or event-specific dynamics, and, critically for reproducibility, the underlying dataset is not publicly available due to licensing restrictions.

The literature suggests that privacy preserving synthetic generation applied to mobility data is understudied. The closest related mobility study operates on aggregated city-scale flows rather than individual trips (Yuan et al. 2026), leaving a clear gap for user-level membership inference on individual mobility records. The observation that privacy risk is uneven — with more active individuals more exposed (Stadler, Oprisanu, and Troncoso 2022; Webster et al. 2021) — has likewise not been applied in this setting. This study addresses these gaps by evaluating user-level membership inference on synthetic origin–destination data from an openly available, sparse dataset, and by examining whether higher-activity users face greater privacy risk.

Chapter 3

Methodology

3.1 The Dataset

For user-level privacy to be assessed it was essential that the dataset had user IDs so mobility patterns can be attributed to an individual, this was a key requirement of the dataset to enable the privacy evaluation. It was also required that the dataset was large enough for a deep learning generation model to generalize well. Aside from these requirements, it helps if the dataset is good quality, representative of true mobility patterns, as well as open source for repeatability. Considering the criteria, the GeoLife dataset was selected as the dataset of choice to base the experiment on.

The GeoLife Dataset

The GeoLife Dataset was collected and published by Microsoft Research Asia. The published dataset consists of GPS trajectories of 182 users, between April 2007 and August 2012. The majority of data is centered in Beijing city, China, with some trips present to other Chinese cities as well as wider cities internationally.

The GPS trajectories cover not only routine movement but also activities such as hiking, shopping, and sightseeing to name a few. The volunteers were motivated to keep GPS loggers on as a participation payment was linked to the extent of their GPS records and trajectories. The dataset has been added to with a subsequent

version update, as detailed in the user guide, detailing corresponding info. (Zheng n.d.)

The dataset consists of trajectories totalling 1.2 million kilometers and over a duration of 48,000+ hours. Over 91% of the data is high frequency dense representation, every 1-5 seconds or every 5-10 meters per point.

The dataset has been used in a number of studies, including assessing user Similarity Based on Location History, predicting points of interests for a user, and predicting transport modes with supervised modelling to name a few papers. Users of the dataset are asked to cite the following linked studies (Y. Zheng, Zhang, et al. 2009) (Y. Zheng, Li, et al. 2008) (Y. Zheng, Xing, and Wei-Ying 2026)

The dataset consists of trajectories made up of many coordinate timestamps, for simplification the dataset was transformed into origin-destination (OD) trips, converting a trajectory of coordinates into a pair of OD coordinates, as part of the preprocessing of data. The following section describes the methodology used to identify and define trips via trip segmentation.

Preprocessing

GPS trajectories were converted into origin–destination trips through stay-point detection, static periods in which a user remained within a small radius for a sustained interval. The detection used a rolling-average centroid for robustness to GPS noise, as well as broadly following the threshold-based approach used by Li et al. (2008); the full procedure and parameters are given in Appendix A. To mitigate GPS noise, trip origins and destinations were then discretised to the H3 hierarchical hexagonal grid at resolution 6, where each cell covers approximately 36 km² (Uber Technologies n.d.). This resolution is fine enough to capture individual travel behaviour, yet coarse enough that locations recur across users — giving the models shared OD flows rather than a sparse field of unique endpoints. Trips beginning and ending in the same cell were discarded, as were all trips falling outside the Beijing municipal boundary (GADM n.d.), restricting the data to a single city. The processed data

yielded 7,739 trips across 154 users, with an average trip being 28.7 minutes and 13.6 km.

A realistic weekly and daily cycle of trip activity was observed. Figure 1 shows the activity of the resulting trips, where it can be observed a clear morning and evening peak, in keeping with other mobility datasets. Furthermore the change in activity over the week can be identified, with the evening peak rush hour observed as later than usual on Fridays, as well as trips on Saturdays and Sundays being more spread out and less concentrated in the morning.

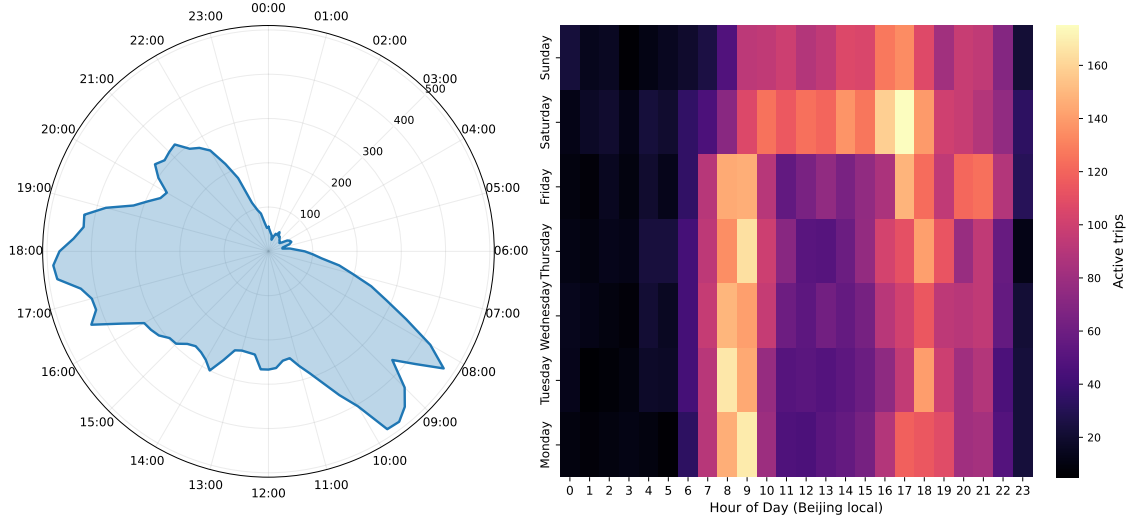


Figure 1: Temporal rhythm of active trips in the real dataset, counted in every interval during which they are in progress. **Left:** number of active trips by time of day in 15-minute buckets. **Right:** active trips by day of week and hour of day. All times are displayed in Beijing local time (UTC+8)

The dataset contains a range of users with a different number of trips, ideal for assessing the effect of activity levels on privacy leakage. The number of trips per users can be seen via the trip count cumulative distribution in Figure 3.

Data Representation A key consideration for deep learning models is the representation format of data; different formats can have an effect on model performance, including its ability to identify real world relations and semantics. For example, discrete spatial representation, requires the model to ‘learn’ spatial awareness during training, i.e. learning that certain grid cells are neighbours.

The spatial representation was kept as continuous coordinates rather than discrete cell IDs to preserve relative spatial awareness for the model: two locations are treated as similar when their latitudes and longitudes are numerically close. Discrete grid cell IDs were considered as an alternative, but raw cell IDs are categorical and carry no inherent order — one cell is no "closer" to an adjacent cell than to any other. Fed to the model directly they would have to be one-hot encoded, a format that treats every cell as equidistant and so discards precisely the spatial relationships the model needs to learn. With a category required for both the origin and destination of each trip, and 253 distinct cells to encode (636 across the wider Beijing region), a one-hot scheme was judged an unnecessarily sparse and high-dimensional way to represent locations whose spatial relationships the model would then have to rediscover from scratch.

Another option is the use of embedding, converting discrete IDs to a vector space based on the resulting data, however such a method requires a sufficiently large dataset for the embedding process to be accurate, this was unsuitable considering the smaller dataset post-processing (7739 trips). The use of grid centroid coordinates represents a simple representation that helps the model understand spatial awareness, in other words, coordinates locations are close to each other if both numeric latitudes and longitudes are close.

Feature Engineering prepares the raw trips into a numerical form the generative models can learn effectively from. Two attributes were derived rather than left for the model to infer. Trip distance was approximated using the haversine distance between the origin and destination centroids coordinates, and trip duration was computed from the start and end timestamps. Supplying these explicitly gives the model the relationship directly, rather than requiring it to learn that origin–destination pairs further apart tend to incur a greater distance and a longer duration. Year and week-of-year information was omitted as the data collection period spans several years unevenly, offering no consistent seasonal patterns. The models can therefore learn from daily and weekly rhythms but not seasonal or longer-term variation.

Gradient updates and training of Neural networks is more stable when inputs are of a comparable scale and centred near zero, therefore each attribute was transformed accordingly. The four centroid coordinates were standardised. Trip distance and duration were log-transformed before z-scoring, as their long-tailed skew (a small number of unusually long trips) would otherwise dominate the scaled distribution. The start hour and day of week were cyclically encoded as sine–cosine pairs, so that the start and end of each cycle are represented as adjacent. The resulting ten-dimensional input vector is summarised in Table 2. All model outputs were inverted through the same transformations back to interpretable trip attributes after generation. Note that the user ID is not included in model inputs as it is only used for the data partitioning and privacy evaluation (as described in Section 3.3).

Table 2: The ten-dimensional continuous feature vector supplied to each generative model.

Feature	Description	Transformation
Origin latitude	Start centroid latitude	Standardised
Origin longitude	Start centroid longitude	Standardised
Destination latitude	End centroid latitude	Standardised
Destination longitude	End centroid longitude	Standardised
Duration	Trip duration (minutes)	Log, then z-scored
Distance	Haversine OD distance	Log, then z-scored
Hour (sine)	Start hour of day, sin term	Cyclical, 24h
Hour (cosine)	Start hour of day, cos term	Cyclical, 24h
Day-of-week (sine)	Start day of week, sin term	Cyclical, 7d
Day-of-week (cosine)	Start day of week, cos term	Cyclical, 7d

3.2 Generative Models

Two deep generative models were implemented, each representing one of the two dominant paradigms in generative modelling: a Variational Autoencoder (VAE), a latent-variable model, and a Wasserstein GAN with gradient penalty (WGAN-GP), an adversarial model. This pairing was chosen to compare the privacy and utility characteristics of architectures with fundamentally different training objectives, rather than to identify a single best-performing model. Both models generated synthetic trips in the same format: the 10-dimensional continuous feature vector.

Generated vectors are inverted back to interpretable trip attributes using the same transformation parameters fitted on the real training data.

Both models share a common training configuration to keep the comparison controlled: the Adam optimiser and a batch size of 64. All stochastic components, weight initialisation and sampling, were anchored to the experiment seed as multiple seed experiments were run as described in the evaluation framework (section 3.3). The hyperparameter exploration process is described in Appendix B and the final hyperparameters used are summarised in Table 3.

Variational Autoencoder

The VAE comprises of a symmetric encoder–decoder pair. The encoder maps the input through two fully connected layers ($10 \rightarrow 64 \rightarrow 32$, each with ReLU activations), from which separate linear layers produce the mean and log-variance of an 8-dimensional approximate posterior. A latent sample is drawn using the reparameterisation trick to permit gradient-based training, as described by (Kingma and Welling 2022); the decoder mirrors the encoder ($8 \rightarrow 32 \rightarrow 64 \rightarrow 10$) to reconstruct the input.

The model is trained to minimise the standard VAE objective, the sum of a reconstruction term and a Kullback–Leibler (KL) divergence term that regularises the latent distribution towards a standard Gaussian prior (Kingma and Welling 2022):

$$\mathcal{L} = \underbrace{\text{MSE}(x, \hat{x})}_{\text{reconstruction}} + \beta \cdot \underbrace{D_{\text{KL}}(q(z | x) \| p(z))}_{\text{regularisation}}.$$

A mean-squared-error reconstruction term was used given the continuous-valued input. The KL weight β was set to 0.1 and ramped linearly from zero over the first 40% of training before being held constant. This annealing schedule mitigates posterior collapse, in which the latent space is ignored early in training. The aggregate posterior diverged from the prior for β choices at low values, where sampling at generation time produced contracted outputs, however a moderate weight of $\beta = 0.1$ improved the posterior–prior match and reduced this effect.

Wasserstein GAN with Gradient Penalty

The WGAN-GP consists of a generator and a critic trained adversarially. The generator maps a 64-dimensional Gaussian noise vector through three hidden layers ($64 \rightarrow 128 \rightarrow 256 \rightarrow 128 \rightarrow 10$), each using batch normalisation and LeakyReLU (slope 0.2) activations, to produce a synthetic feature vector. The critic mirrors this capacity ($10 \rightarrow 128 \rightarrow 256 \rightarrow 128 \rightarrow 1$) and outputs an unbounded real-valued score rather than a probability.

Unlike a conventional GAN, the critic is trained to approximate the Wasserstein distance between the real and synthetic distributions (Arjovsky, Chintala, and Bottou 2017), which provides smoother gradients and reduces the mode collapse and training instability associated with the original GAN formulation. The 1-Lipschitz constraint required by this formulation is enforced through a gradient penalty (Gulrajani et al. 2017): the squared deviation of the critic’s gradient norm from unity, evaluated at points interpolated between real and synthetic samples, is added to the critic loss with weight $\lambda_{gp} = 10$. The critic was updated three times per generator update, and the standard Adam configuration for this method ($\beta_1 = 0.5$, $\beta_2 = 0.9$) was used.

One architectural detail follows directly from the gradient penalty: the critic uses layer normalisation rather than batch normalisation. Because the penalty is computed per sample, batch normalisation — which couples samples within a batch — would invalidate the per-sample gradient constraint. Layer normalisation normalises each sample independently and is therefore the appropriate choice. The generator, which is not subject to this constraint, retains batch normalisation.

Table 3: Final hyperparameters for the generative models.

Setting	Description	Value
<i>Shared</i>		
Optimiser	Adam	—
Batch size	Samples per batch	64
<i>VAE</i>		
Latent dimension	Size of latent space	8
Learning rate	Adam step size	1×10^{-3}
β	Final KL weight	0.1
KL annealing	Warm-up fraction of epochs	0.4
Epochs	Training epochs	80
<i>WGAN-GP</i>		
Noise dimension	Generator input dimension	64
Hidden dimension	Base width of generator/critic	128
Learning rate	Adam step size	1×10^{-4}
λ_{gp}	Gradient penalty weight	10
Critic steps	Critic updates per generator update	3
Adam (β_1, β_2)	Adam momentum terms	(0.5, 0.9)
Epochs	Training epochs	100

3.3 Evaluation Framework

The evaluation framework is clearly split into two sections, an assessment of utility, for which the similarity to the real dataset is included as a proxy. Secondly, the privacy assessment and how well it protects individuals from privacy leak. Ideally a synthetic dataset would score well on both, maintaining strong utility while obtaining a strong privacy score.

A Comparison Baseline – To provide a consistent point of reference, all evaluation metrics are reported relative to a bootstrap baseline model. The bootstrap generator creates synthetic samples by randomly resampling trips from the training dataset (with replacement), thereby preserving the empirical data distribution without performing any pattern learning or generalisation. For clarity - future reference of "generative models" refers to the two deep learning generators used (VAE and WGAN-GP), the "benchmark" refers to Bootstrap resampling model used as a comparison, where "all models" refers to both groups.

Comparing each deep generative model against this benchmark helps distinguish improvements that arise from learned representations rather than simple replication of the original data. As a result, it provides an approximate upper bound for distributional utility, while simultaneously representing a worst-case baseline for privacy due to its direct reuse of training examples. In terms of privacy, the bootstrap model provides a useful reference because it represents the extreme case of direct data reuse, against which reductions in information leakage can be assessed. This enables the privacy and utility characteristics of different models to be interpreted relative to a simple, transparent baseline, while enabling the explicit assessment of the privacy-utility trade-off.

Repeated seed runs were carried out to obtain more accurate metrics, as well as obtaining the variability and confidence range. Five seed runs were carried out. Any randomised processes (e.g. data splits, deep learning generation seeds) were anchored to the experiment seed run for reproducibility.

Utility Metrics

To answer RQ1 — whether the generative models reproduce real origin–destination (OD) mobility patterns — the utility evaluation was conducted along two lines. The first is a set of distributional-similarity metrics that measure how closely the synthetic data matches the real data across its spatial, temporal and continuous attributes. The second is a single downstream forecasting task that tests whether the synthetic data preserves enough structure to stand in for real data on a realistic predictive problem. Every metric was computed independently for each of the five random seeds, for each metric the mean and standard deviation was reported across seed runs.

The Jensen–Shannon distance (JS) was used as one of the main similarity metrics. It is calculated on discrete attributes, for example how often a trip starts from a spatial cell (origin), or how often it departs from an hour in the day. The synthetic data and real data are put into discrete bins corresponding to that attribute,

from this two normalised histograms can be formed and compared. As described by Lin (1991), the JS divergence can be computed using base-2 logarithms, and then square-rooted to give the JS distance, bounded by $[0, 1]$. A distance close to zero represents close similarity, whereas a larger distance represents low similarity. The JS distance was calculated for several attributes: the origin cells of trips, the destination cells of trips, and the origin–destination pairs (OD-flow) — all discretised to H3 resolution-6 cells. The JS distance for origin and destination are also called the marginals, where as the OD-flow is the joint distribution, representing a sparser distribution of origin-destination cell pairs.

For the temporal attributes, the JS distance was calculated for the hour of day (discretised into 24 bins) and the day of the week (discretised into 7 bins). These capture whether a model reproduces the rhythm of travel — morning and evening commute peaks, and the weekday/weekend contrast — independently of where the trips occur.

OD-flow coverage was calculated as proportion of distinct real OD pairs that appear at least once in the synthetic data, using the same discrete cells. Unlike the JS distance, it does not consider the number of trips but just whether at least one match exists. 1 indicates full coverage, whereas 0 represents no coverage of OD pairs. Note a model could potentially achieve deceptively low distance (close distribution) by reproducing a few high-frequency flows well while having low coverage, never generating the long tail of rarer OD trips. Because of this, the OD-flow JS distance was considered the principal metric for whether a model replicated similar mobility patterns.

The following three comparison statistics operated on raw coordinates (origin latitude and longitude, destination latitude and longitude), treated as continuous variables sidestepping any choice of a spatial grid. The final metric displayed is the average across the four coordinate values.

The Wasserstein distance , specifically the 1-Wasserstein distance, was calculated on the coordinates of origin-destination values. Also known as the Earth

Mover’s distance, it represents the ‘work’ needed to reshape the coordinate distribution of the synthetic dataset to the real dataset distribution, both distributions were first standardized to the real data distributions. A lower distance score represents a closer distribution. (Rubner, Tomasi, and Guibas 2000)

The Kolmogorov–Smirnov (KS) statistic was also calculated on the coordinate distributions, representing the largest vertical gap between the real and synthetic cumulative distributions of the coordinate space (Massey 1951). Again, a lower score represents closer similarity.

The Spread Ratio is the standard deviation of the synthetic coordinate distribution divided by that of the real coordinate distribution, averaged across the four coordinate values. It is a simple variance preservation check, where a value of 1.0 indicates the synthetic data has the same spatial dispersion as the real data, a value below 1.0 indicates it is under-dispersed (spatially compressed), and a value above 1.0 indicates it is over-dispersed (spatially stretched distribution).

Correlation difference - Faithful synthetic data should preserve not only each attribute’s marginal distribution but also the relationships between attributes — for instance, that longer trips tend to take more time. Therefore the Pearson correlation matrix was calculated over the trip duration, distance and the four OD coordinates for the real and synthetic data. The elementwise absolute difference of the upper triangles of the matrix was calculated to obtain a mean difference. Lower value indicates a closer correlation preservation, while a large value signifies the wrong joint relationships, even if the synthetic data has the right marginals.

Downstream utility - Lastly, a forecast task was applied to assess whether the synthetic data could be useful substitute to real data for a practical task. For this the Train-on-Synthetic Test-on-Real evaluation protocol was followed (TSTR), as proposed by Esteban, Hyland, and Rättsch (2017). For this the forecast task was defined as inferring a trip’s destination cell (H3 resolution-6) from its origin coordinates, and trip duration and trip distance. The classifier used was a random

forest classifier (200 trees, balanced class weights). Two identical classifiers were trained: one on the synthetic data and another on a real training split. Both were then evaluated on the same held-out split of real trips and the F_1 score for each was recorded. The classifier trained on the real data (TRTR) provides a ceiling, a realistic best score achievable. The final downstream utility of the synthetic dataset was reported as the TSTR/TRTR ratio. A ratio of 1.0 means a model trained purely on synthetic data is as useful as one trained on real data for the task, whereas values below 1.0 quantify the utility lost. Expressing the result as a ratio normalises out the intrinsic difficulty of the task, so the figure reflects the quality of the synthetic data rather than how hard the prediction task happens to be.

Privacy Attacks

Two methods were used to assess privacy leakage, the first being the origin-destination reproduction rate (near-copy rate), and the second being a Membership Inference Attack assessment.

Near-copy rate - The near-copy rate measures how often a synthetic generator reproduces a real training trip. For each synthetic trip, the Euclidean distance to its nearest neighbour among the real training trips is computed over the four origin-destination coordinates, and the near-copy rate is the proportion of the synthetic trips whose nearest-neighbour distance falls below the distance threshold ($\varepsilon < 0.012$ in the standardised coordinate space) — chosen to correspond roughly with the H3-cell resolution. A high near-copy rate indicates that the model is effectively copying training records rather than generating novel ones, which is a direct privacy risk; a rate near zero indicates that the origin-destination pairs for synthetic trips are genuinely new.

Membership Inference Attacks (MIA) were used as the main assessment metric for privacy, specifically MIA at the user level; this attack attempts to determine whether a given user was present in the training set of the generator that produced the synthetic dataset. A high attack success rate indicates that the user’s travel

patterns have been strongly memorised by the generator.

This evaluation follows the shadow-model paradigm introduced by Shokri et al. (2017), in which multiple shadow models are trained on known member/non-member (IN/OUT) splits to produce labelled examples, on which an attack classifier learns to distinguish membership. Stadler, Oprisanu, and Troncoso (2022) adapted this process to generative models, where the attacker infers membership from a sampled synthetic dataset by first reducing it to a fixed length feature vector before classification. The approach taken here follows the same structure, applied at the user rather than record level and adapted to the mobility domain.

Each user is represented via a feature vector that compares that user’s real trips against the synthetic dataset as a whole. Features could be as basic as differences in the mean or variance of trip attributes, or more targeted comparisons such as nearest-neighbour distances in OD space. To train the attack classifier, ten shadow generators were trained on known IN/OUT user splits, and the classifier learns to identify which combinations of features indicate membership. Once trained, it is applied to the target generator’s synthetic output to assess membership for the held-out evaluation users. Figure 2 shows the user splits and the attack-classifier training flow.

A shadow generator is essentially a replica of the main generator with the exact same architecture, as well as model and training parameters, but anchored with a different seed and trained on a different dataset. These multiple shadow generators help provide enough training examples for the attack classifier to ensure effective performance during final MIA evaluation.

Users labelled as shadow users are used for both training of the main generator as well as the MIA attack classifier, Eval In users are only used in generation of main synthetic users, whereas Eval Out users are only for the final evaluation.

To ensure a realistic evaluation, users employed in the training of the shadow generators, and therefore the attack classifier, are excluded from the final MIA assessment. This prevents information leakage between attack training and evaluation, ensur-

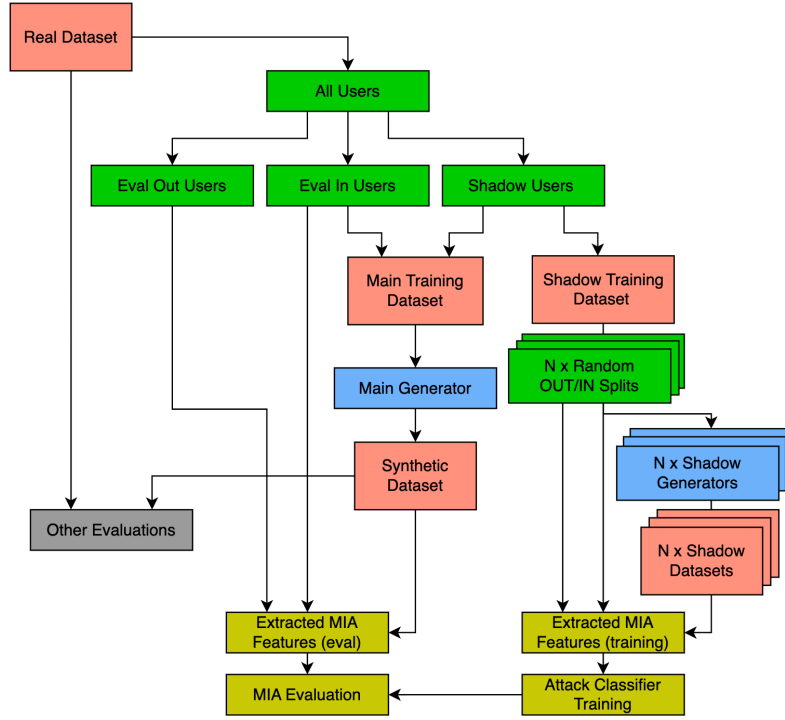


Figure 2: Overview of the evaluation pipeline, emphasising the user partitioning underpinning the membership-inference attack (partition splits are stated in Table 5. Colours: red — datasets; green — partitioned users (member/non-member); blue — synthetic-data generators; yellow — MIA features and evaluation.

ing that attack performance reflects the ability to infer membership for previously unseen users rather than users whose characteristics were already observed during attack development.

Attack features are created by calculating comparison statistics between the set of trips of the real user in question, and that of the synthetic dataset trips. These set of features are then fed into the classifier for inference on whether that user was present in the training of the synthetic dataset. The thirteen per-user features are grouped into five families and summarised in Table 4. The majority of the features are derived from the distribution of Nearest-neighbour distances — the Euclidean distance from each of the user’s trips to its closest synthetic trip in origin–destination (OD) coordinate space, giving one distance per trip. The first two families are summaries of this distance array.

The feature set is deliberately weighted towards locational signals as user’s member-

Table 4: Membership-inference attack features.

Family	Features and description
NN OD-distance summary (7 features)	Calculated from the NN distance array: min , p01 , p05 , p25 , median , mean , std
Close-match rates (2)	Threshold proportions from the same NN distance array: close_trip_rate (<0.025) and very_close_trip_rate (<0.012) — share of the user’s trips with a near-identical synthetic neighbour
OD overlap (1)	od_overlap : fraction of the user’s distinct H3 OD cell-pairs present in the synthetic set (cell resolution - 6)
Aggregate similarity (2)	Closeness of the user’s <i>mean</i> value to the synthetic-set mean: distance_similarity (trip length) and duration_similarity (trip duration)
Support (1)	n_target_trips : the user’s trip count

ship fingerprint in mobility is carried principally by where they travel to and from — locations personal to individuals such as home and work cells are most likely to expose membership. Not all attack features will be as useful as each other for determining membership, however the attack classifier should be able to determine which are useful and when a certain combination of features indicated membership or not. The classifier is therefore strongly coupled with the chosen attack features.

These distribution statistics that form the attack features become unstable for users with very few trips, in such cases the membership signal is dominated by sampling noise rather than genuine memorisation. For this reason, users with fewer than ten trips are excluded from the MIA evaluation process. Although these MIA-excluded users are not shown in Figure 2, conceptually they are taken from the real dataset and are included into the main training dataset only so models have additional training examples of real mobility trips. After removing these ineligible users, 85 users remain to form the MIA process – divided between Eval In, Eval Out, and Shadow Users.

Train-test splits - As illustrated in the diagram, the membership inference evaluation rests on a user-level partition of the data, constructed independently for each of the five seeds. Splitting at the user level rather than the trip level ensures that all of a given user’s trips fall on the same side of the membership boundary, so that membership is defined per user. 20% of users are held out entirely as non-members (*Eval Out Users*) used for the final MIA evaluation, the remaining 80% of users are used for the training of main synthetic generator models.

For the MIA assessment to work effectively a trained attack classifier is required. As shown in Table 5 the remaining 80% of users are split equally into a *Eval In* pool, those used as In members for final evaluation, as well as a *Shadow User* pool — those exclusively used to train the shadow generators that produce labelled data for the attack classifier to train on. This separation avoids contamination between training and evaluation — whereas In Eval users required for final MIA evaluation are not used for classifier training.

Table 5: User-level partitions of the 85 MIA-eligible users; those with very low activity (<10 trips) are not MIA eligible and are excluded from any inference evaluation

Split	% of eligible	Purpose
Eval Out	20%	Non-members: held out from the target generator entirely. Form the negative (non-member) class against which the attack is tested.
Eval In	40%	Members: their real trips are in the target generator’s training set. Form the positive (member) class against which the attack is tested.
Shadow	40%	Used to generate the shadow datasets for classifier training. Each of the ten shadow generators trains on a random half of these users, with the other half held out, producing known member/non-member examples to train the attack classifier. Also used in main-generator training but not final evaluation of the main-generator.

To ensure a well trained classifier with enough data, 10 different shadow generators are formed (each with a different seed) to produce 10 different shadow datasets. Each shadow generator is trained on a random half of the shadow users, with the other half held out so that the attack classifier learns from known in/out labels

before being applied to the final MIA evaluation users.

Derived Attack Metrics - The MIA classifier attempts to determine whether a given user’s data was part of the model’s training set or not, outputting a probability score. The probability score along, with the known membership of the user, was used to derive four metrics representative of the attack success. **Balanced accuracy** uses a 0.5 probability threshold to determine the classifier’s binary assessment. It then averages the true-positive and true-negative rates (TPR & TNR); Unlike raw accuracy, it is not inflated by any imbalance between the number of members and non-members. A balanced accuracy score of 0.5 indicates random guessing from the classifier and an inability to infer membership, whereas 1.0 represents a perfect attack rate.

Membership prediction requires the choosing of a probability threshold above which a user is labelled as a member, the conventional choice of 0.5 is somewhat arbitrary, a different threshold would trade true positives against false positives differently. Rather than committing to a single threshold, the attack can be characterised across all thresholds with the ROC (receiver operating characteristic) curve which plots the true-positive rate against the false-positive rate as the decision threshold on the membership probabilities is varied across its full range (Carlini et al. 2022). The **AUC** is the area under the attack’s ROC curve, representing a more general summary of attack success at different thresholds. Similarly to accuracy, an AUC score of 0.5 indicates no better than random separation, while 1.0 is a perfect attack.

Maximum advantage, also known as Yeom advantage, is the largest value of TPR-FPR difference (true-positive rate – false-positive rate), attained at any threshold, capturing the best-case separation available to an adversary when free to choose its operating point. This corresponds to the greatest vertical distance between the ROC curve and the ROC diagonal (representing random chance) — the single operating point at which the attacker best separates members from non-members. An advantage of 0 denotes no advantage over random guessing and 1 denotes perfect attack rate at that operating point. (Yeom et al. 2018)

The final metric is the **true-positive rate at a fixed false-positive rate** which asks a more targeted question: if the attacker is only allowed to be wrong a small fraction of the time (falsely accusing at most 1%, 5% or 10% of non-members), how many real members can it still correctly identify. Therefore a value at chance corresponds to the true-positive rate equalling the false-positive rate. This matters because the other summary metrics average the attack’s performance across all operating points, including ones an adversary would never use, whereas a real attacker would want to identifying membership confidently. The true-positive rate at a low false-positive rate captures this worst-case threat, and is the operating point emphasised in the membership inference literature. (Carlini et al. 2022)

Splitting the users into IN/OUT attack assessment groups results in a smaller number of users in each, making statistical calculations more unstable, therefore all MIA associated privacy metrics were pooled across experimental seed runs for better metric stability. Furthermore, from this collated distribution a 95% percentile-bootstrap confidence interval was extracted — 2000 samples with replacement were used to infer the interval, representing the metric’s 2.5th–97.5th percentile. Note that this bootstrapping is not to be confused with the benchmark generation model, however the two use the same technique of sampling with replacement, whereas one is used to form a mobility dataset and another is used to obtain a confidence interval.

In contrast, the non-MIA associated evaluation metrics are more stable due to fewer dataset partitions. Therefore all other metrics are reported with a simple ± 1 standard deviation, representing the variability between the seed runs.

Assessing by Activity Level

To assess the privacy protection by activity level of users (RQ3), users were grouped into three groups based on the number of trips each user has. To ensure a balanced evaluation, activity groups were partitioned so the number of users in each activity group was equal, as shown in Table 6.

Users with less than 10 trips are excluded from any MIA evaluation so are not

included in the MIA group stratification by activity level. Note these activity group sizes are fixed and are identical across all five seed experiments. Figure 3 indicates the full distribution of the number of user trips, showing that a small number of users undertake a large number of trips, as shown by the plot’s long tail.

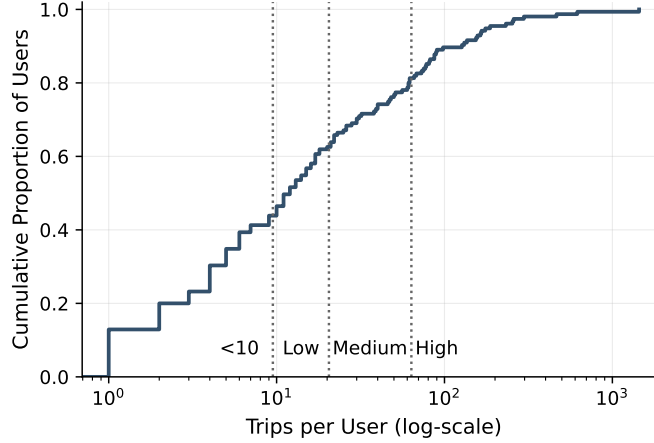


Figure 3: Empirical cumulative distribution of trips per user across all users in the real dataset. Dotted vertical lines mark the boundaries of the three activity groups used in RQ3, as well as those excluded (<10 trips).

Table 6: User activity group, defined by total trip count per user.

Group	Trips/user	Description	Users
MIA Excluded	≤ 9	Too few trips for reliable membership inference (non-eligible)	67
Low	10–20	Lower third of eligible users	29
Medium	21–63	Middle third of eligible users	29
High	≥ 64	Upper third of most active eligible users	29
Total			154

All train-test splits in the MIA assessment, as shown in Table 5, were performed with activity group stratification, each split has a appropriate proportion of low/medium/high activity users (e.g. approximately 20% of high-activity users are in the Eval Out group). Note the number of activity-level users in each group after stratification is approximate, as data splits are not always divisible by 3 equal groups (activity level groups).

Chapter 4

Results

4.1 RQ1: Utility

Table 7 shows the utility metrics for each model, including the variability across seed runs.

	VAE	WGAN-GP	Bootstrap
Origin JS Distance	0.852 ± 0.007	0.846 ± 0.008	0.154 ± 0.068
Destination JS Distance	0.851 ± 0.003	0.843 ± 0.009	0.159 ± 0.066
OD-flow JS Distance	0.672 ± 0.027	0.637 ± 0.035	0.308 ± 0.077
OD-flow Coverage	0.427 ± 0.018	0.372 ± 0.013	0.673 ± 0.050
Hour-of-day JS Distance	0.154 ± 0.021	0.149 ± 0.023	0.058 ± 0.024
Day-of-week JS Distance	0.116 ± 0.010	0.117 ± 0.014	0.013 ± 0.006
Correlation Difference	0.071 ± 0.013	0.052 ± 0.019	0.030 ± 0.010
Wasserstein Distance	0.167 ± 0.002	0.120 ± 0.005	0.018 ± 0.004
Spread Ratio	0.804 ± 0.011	0.908 ± 0.024	1.007 ± 0.012
Kolmogorov-Smirnov Distance	0.166 ± 0.012	0.157 ± 0.014	0.219 ± 0.017
Forecast Task (TSTR)	0.271 ± 0.045	0.251 ± 0.038	0.987 ± 0.042

Table 7: RQ1 utility results (mean $\pm$ standard deviation over five seed runs). JS Distance denotes the Jensen-Shannon distance between the corresponding real and synthetic marginal distributions. Mean Wasserstein Distance is computed on features standardised using statistics from the real data

Figure 4 shows spatial and temporal Jensen-Shannon (JS) distances between synthetic distributions and real data.

The deep learning generative models show significant spatial divergence, with both Origin-cell and destination-cell consistently exceeding 0.8 for both the VAE and WGAN-GP. In contrast the Bootstrap baseline shows a much smaller divergence of 0.154 and 0.159 respectively. The joint OD-flow distance follows the same pattern with generative models showing a significantly higher divergence than the baseline: 0.672 for the VAE, 0.637 for WGAN-GP, in comparison to 0.308 for the Bootstrap.

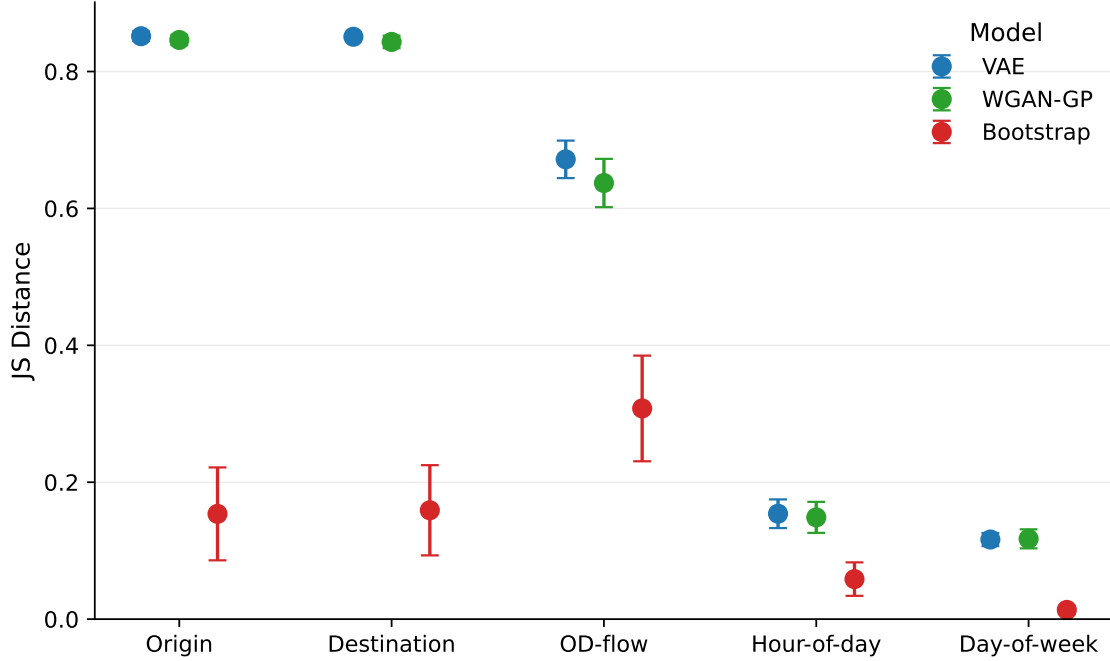


Figure 4: Distributional Fidelity Across Spatial and Temporal Mobility Distributions. Jensen-Shannon (JS) distances between the real and synthetic distributions for origin cells, destination cells, OD-flow pairs, hour-of-day, and day-of-week distributions. Points indicate the mean JS distance across five seed runs and error bars denote ± 1 standard deviation.

The spread ratio indicates that the VAE contracts the spatial distribution relative to the real data (0.804), while WGAN-GP remains closer to the real dispersion (0.908). These values are consistent with the spatial maps in Figure 5, showing the VAE concentrates trips into a more compact central mode around Beijing’s urban core, whereas WGAN-GP produces a broader distribution covering more of the real support. Both deep models also place generated synthetic trips in new locations, including locations beyond the administrative boundary of Beijing where real trips were omitted from the dataset.

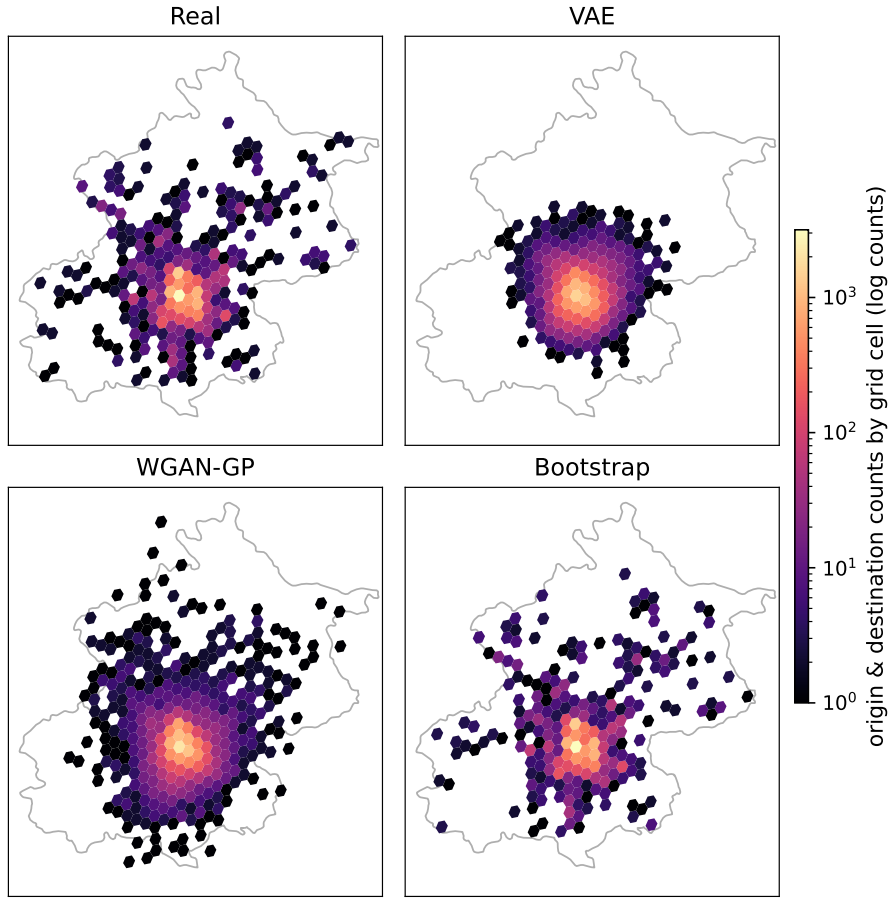


Figure 5: Spatial distribution of OD trips for the real training-split data and each generative model (seed 111). Each panel maps the log count of origin and destination points falling within H3 resolution-6 hexagonal grid cells, on a shared colour scale across all panels. The grey outline is the GADM 4.1 administrative boundary of Beijing municipality. See Appendix C for the equivalent grids across all five seed runs.

It can be seen that the generative models reproduced the temporal patterns better than the spatial travel behaviours. Weekly patterns were replicated well with low JS distance for day-of-week: 0.116 (VAE), 0.117 (WGAN-GP), and 0.013(Bootstrap), showing close alignment with number of trips per day and hour.

Daily trip patterns were also replicated closely with low distances for Hour-of-day with JS distances of 0.154 (VAE), 0.149 (WGAN-GP), and 0.058 (Bootstrap). This aligns with Figure 6 showing similar hourly peaks to the real data for each model. The Bootstrap tracks the real data most closely with the deep learning models deviating mainly in inter-peak hours.

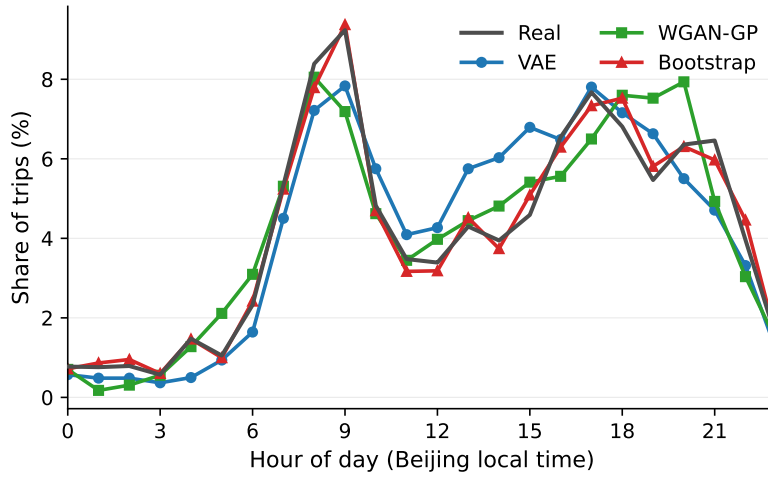


Figure 6: Distribution of trip start times by hour of day for the real data and each generative model (seed 111), expressed as the share of trips falling in each hourly bin (Beijing local time, UTC+8).

Downstream predictive utility separates the methods most sharply (Figure 7). The train-on-synthetic, test-on-real (TSTR) ratio for destination prediction is 0.271 for the VAE and 0.251 for WGAN-GP, against 0.987 for the Bootstrap, which almost sits at the maximum real-data ceiling.

A classifier trained on synthetic data from the deep-model therefore recovers roughly a quarter of the performance obtainable from real data, whereas one trained on Bootstrap data is near-equivalent to training on the real set.

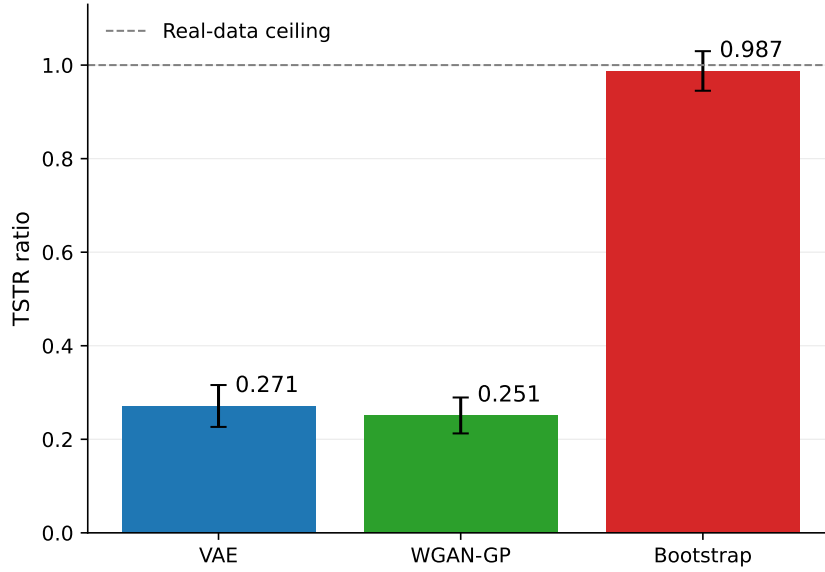


Figure 7: Forecast task utility (Train-on-Synthetic, Test-on-Real). The TSTR ratio is defined as the macro-F1 score obtained when training on synthetic data and testing on real data, divided by the performance obtained when both training and testing on real data. Bars show the mean TSTR ratio across five seed runs and error bars denote ± 1 standard deviation. The dashed line at 1.0 represents the real-data performance ceiling, with values closer to one indicating better downstream utility preservation.

4.2 RQ2: Privacy Protection

The near-copy rate was shown to be around 1% for the deep learning models, as shown in Table 8. This represents the proportion of trips that closely align to one of the trips in the real dataset, therefore further clarifying the deep learning models ability to alter or generate new trips. The benchmark Bootstrap model showed a 100% near-copy rate as expected, due to its simple sampling technique from the real dataset.

The results of the membership inference attack, as shown in Table 8, indicates that no measurable signal from either deep generative model was detected with attack accuracy scores around 0.5, representing random guessing.

The membership inference attack produces a score for each user, expressing the attacker’s confidence that the user’s data was used in training. Sweeping a decision threshold over these scores yields the ROC curve. The AUC score for the VAE

	VAE	WGAN-GP	Bootstrap
Users Evaluated	85	85	85
Balanced Accuracy	0.515 [0.460, 0.571]	0.474 [0.417, 0.529]	0.692 [0.629, 0.757]
AUC	0.497 [0.433, 0.567]	0.503 [0.432, 0.575]	0.750 [0.673, 0.823]
Maximum Advantage	0.050 [0.032, 0.177]	0.070 [0.020, 0.206]	0.446 [0.350, 0.577]
TPR @10% FPR	0.114 [0.062, 0.194]	0.103 [0.038, 0.184]	0.187 [0.078, 0.386]
TPR @5% FPR	0.069 [0.039, 0.135]	0.059 [0.023, 0.116]	0.081 [0.018, 0.237]
TPR @1% FPR	0.000 [0.000, 0.087]	0.030 [0.010, 0.070]	0.013 [0.002, 0.074]
Near-copy Rate	0.009 $\pm$ 0.001	0.010 $\pm$ 0.002	1.000 $\pm$ 0.000

Table 8: RQ2 privacy results. MIA-associated metrics are pooled across seed runs and reported with 95% cluster-bootstrap confidence intervals. Users Evaluated denotes the number of distinct users contributing to each pooled estimate. Near-copy Rate is reported as mean $\pm$ standard deviation over seed runs.

and WGAN-GP was 0.497 and 0.503 respectively, corresponding to a near random attack success (AUC = 0.5). Also evidenced in Figure 8, showing the ROC curves for both deep-learning models lying along the diagonal. This low attack success is also reflected in the small attacker advantage for the deep learning models (0.05 and 0.07 for the VAE and WGAN-GP respectively).

In contrast the Bootstrap baseline is much more distinguishable from random chance, with an AUC of 0.750 and a curve bowing towards the top-left, indicating that the attack was successfully distinguishing members from non-members. Consequently the benchmark Bootstrap shows a high maximum advantage (0.446).

Despite there being large confidence intervals (Figure 9), the AUC scores for the baseline and generative models are clearly separable with no overlap in confidence intervals.

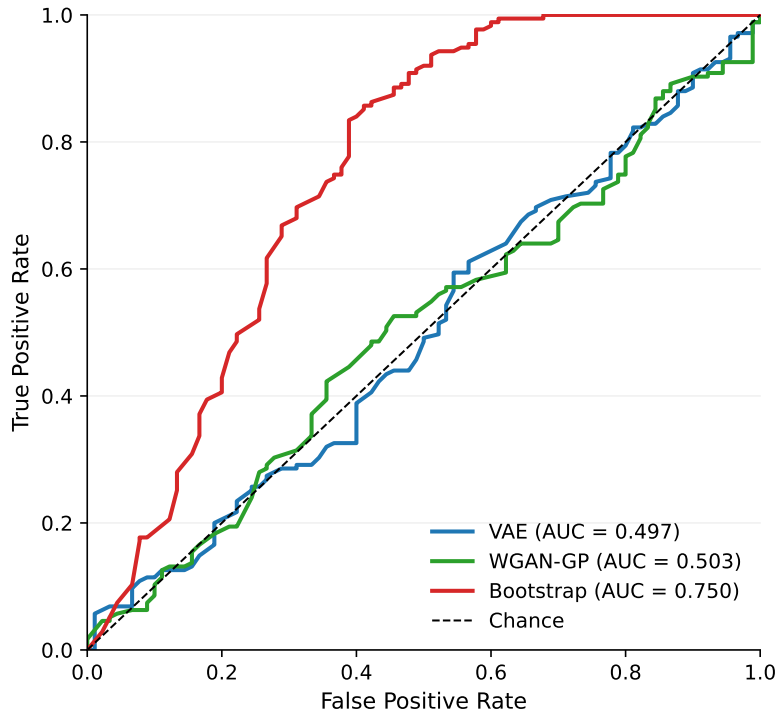


Figure 8: Membership inference receiver operating characteristic (ROC) curves pooled across all seed runs. The area under the curve (AUC) quantifies the attacker’s ability to distinguish training members from non-members, with the dashed diagonal corresponding to random guessing (AUC = 0.5). Curves further above the diagonal indicate greater privacy leakage.

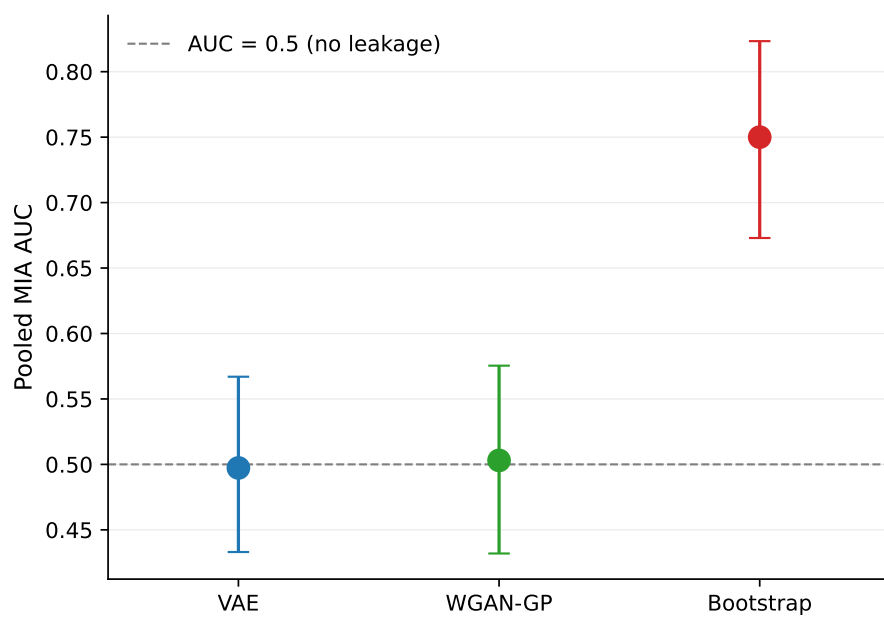


Figure 9: Membership inference attack (MIA) leakage by model. Points indicate pooled MIA AUC estimates and error bars denote 95% cluster-bootstrap confidence intervals obtained by resampling users. The dashed line at $\text{AUC} = 0.5$ indicates no membership leakage, with values closer to 0.5 corresponding to greater privacy preservation.

4.3 RQ3: Privacy by Activity Level

Table 9 reports some of the key privacy metrics broken down by user activity group, AUC is shown as a useful summary of the overall ROC curve, whereas the true positive rate at a fixed 10% false positive rate (TPR @ 10% FPR) characterises the attack success at the ROC curve location where membership inference attacks are typically evaluated (Carlini et al. 2022). As previously, all MIA values are pooled across the five seeds, with 95% cluster-bootstrap confidence intervals obtained by resampling users within each group.

Model	Activity Group	Users	AUC	TPR @ 10% FPR	Near-copy Rate
VAE	Low (10–20)	28	0.580 [0.440, 0.716]	0.186 [0.057, 0.379]	0.004 $\pm$ 0.001
VAE	Medium (21–63)	29	0.530 [0.460, 0.610]	0.263 [0.113, 0.362]	0.005 $\pm$ 0.001
VAE	High (64+)	28	0.392 [0.273, 0.517]	0.102 [0.000, 0.193]	0.008 $\pm$ 0.001
WGAN-GP	Low (10–20)	28	0.578 [0.468, 0.692]	0.220 [0.024, 0.400]	0.006 $\pm$ 0.002
WGAN-GP	Medium (21–63)	29	0.490 [0.382, 0.598]	0.193 [0.029, 0.327]	0.007 $\pm$ 0.002
WGAN-GP	High (64+)	28	0.460 [0.320, 0.612]	0.068 [0.016, 0.196]	0.010 $\pm$ 0.002
Bootstrap	Low (10–20)	28	0.729 [0.618, 0.858]	0.237 [0.019, 0.586]	0.483 $\pm$ 0.040
Bootstrap	Medium (21–63)	29	0.770 [0.652, 0.874]	0.316 [0.036, 0.569]	0.613 $\pm$ 0.011
Bootstrap	High (64+)	28	0.746 [0.582, 0.895]	0.237 [0.038, 0.623]	0.935 $\pm$ 0.025

Table 9: RQ3 privacy by activity group. MIA-associated metrics are pooled within each activity group and reported with 95% cluster-bootstrap confidence intervals. Users denotes the number of distinct users contributing to each pooled estimate. Near-copy Rate is reported as mean $\pm$ standard deviation over seed runs.

Figure 10 shows the MIA AUC by user activity group. The Bootstrap baseline remains above the chance line in all three groups (intervals excluding 0.5) but shows no clear monotonic trend across activity levels in this metric.

For both the generative models, the point estimates decline slightly as activity increases — an opposite gradient direction to that predicted by RQ3. The high-activity VAE estimate falls significantly below 0.5, consistent with sampling noise around chance rather than indicating any further privacy gain. Furthermore, the 95% confidence intervals for the generative models spans the 0.5 chance line, reflecting the smaller number of users available to resample in each group. Because the intervals overlap substantially and include 0.5, no difference in attack accuracy across activity groups can be established for the generative models.

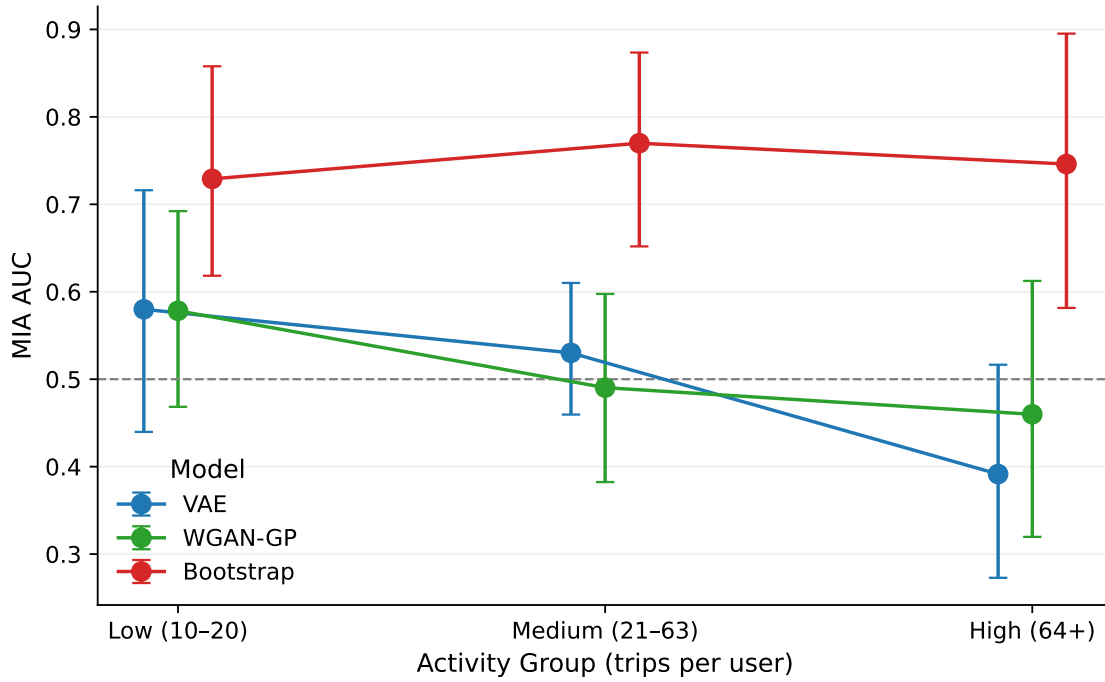


Figure 10: Membership inference attack (MIA) leakage by user activity group. Activity groups correspond to the number of trips contributed by each user. Points indicate pooled MIA AUC estimates and error bars denote 95% cluster-bootstrap confidence intervals obtained by resampling users within each activity group. The dashed line at $AUC = 0.5$ indicates no membership leakage.

Finally, the near-copy rate per group was analysed, measuring the direct record reproduction independently of any trained attacker. The near-copy rate was observed to monotonically increase as activity level increases, consistent across all generative models as well as the benchmark. The WGAN-GP was shown to have greater reproduction variability than the VAE, with a larger standard deviation between experiments. Although the WGAN-GP resulted in some overlap of error-bars between activity group, the Bootstrap and VAE models resulted in negligible or no overlap at all.

Although a similar activity level trend was observed for all models, the Bootstrap replicates significantly more records, as seen by the different axis needed in Figure 11. The Bootstrap has a near-copy rate of 48% for low activity users, increasing up to 94% for the high activity users, evidencing the increasing likelihood of at least one user trip matching a trip in the real dataset despite train-test splits carried.

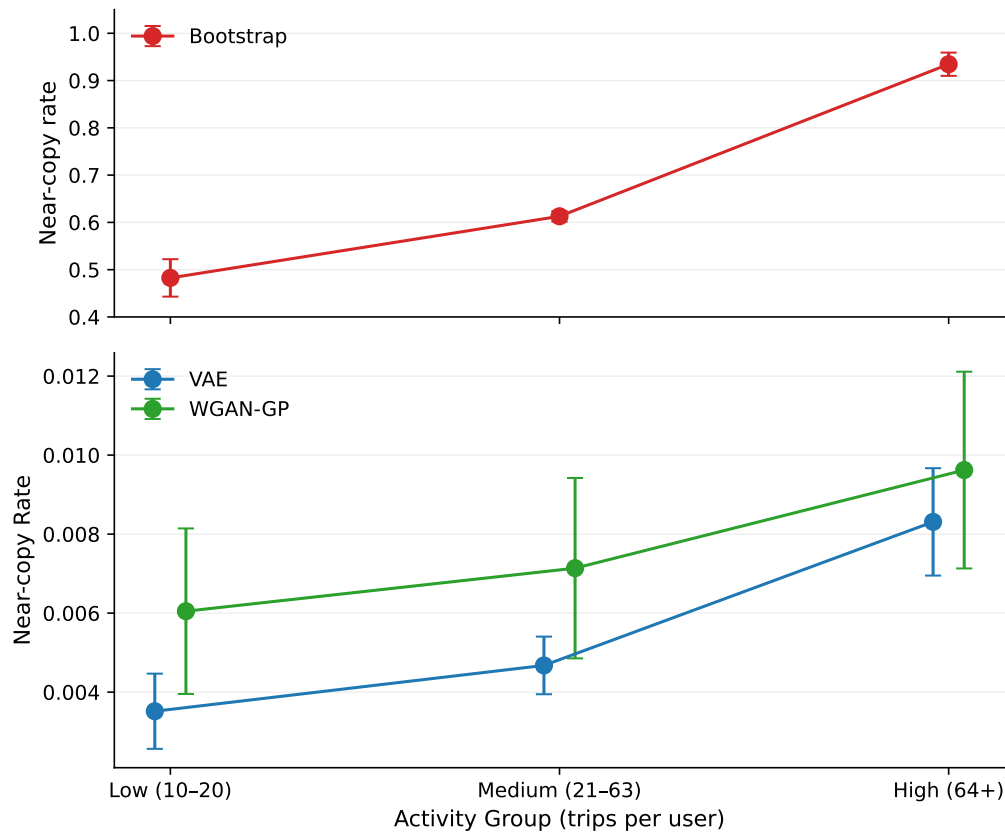


Figure 11: Near-copy rate by user activity group. The upper panel shows the Bootstrap control and the lower panel shows the deep generative models. Points indicate the mean near-copy rate across five seed runs and error bars denote ± 1 standard deviation. Higher values indicate a greater tendency to reproduce records closely resembling those in the training data.

Chapter 5

Discussion

5.1 Interpretation

The generative models reproduced the high-level temporal behaviour markedly better than the spatial behaviour. Temporal marginals were captured closely, with hour-of-day and day-of-week JS distances of roughly 0.12–0.15 for both models, as also indicated by the replication of the morning and evening peaks in Figure 6. Whereas the spatial marginals diverged sharply, exceeding 0.8 for both origin and destination cells.

The same spatial weakness surfaced in the downstream task, which asked whether a destination cell could be predicted from a trip’s origin, duration and distance. Here the separation is starkest: a classifier trained on the synthetic data recovered only around a quarter of the performance achievable on real data (TSTR ratio 0.25–0.27), in comparison to 0.99 for the bootstrap. A synthetic dataset that is temporally realistic but spatially diffused is of limited use for any task that relies on close spatial alignment.

This contrast most likely reflects the differing complexity of the two attributes. Temporal patterns are low-dimensional and strongly cyclic, so the distribution can be approximated more easily from fewer examples. Spatial behaviour, by contrast, is spread across a high-cardinality grid of possible origins and destinations, with far

more structure to learn and far less data per cell.

One result that appears counter-intuitive is that the OD-flow JS distance is lower than the individual origin and destination distances for the generative models (Figure 4), despite the joint OD-flow distribution being harder to match than either marginal. This most likely reflects that the sparsity of the joint space dilutes their already-large placement error, pulling the OD-flow distance below their marginals. Additionally, the JS distances are computed over different distributions and dimensionality, therefore they are not directly comparable in magnitude, even if resulting in a distance score on the same $[0,1]$ scale.

The two generative models differ in how they treat spatial spread, this can be considered reflective of their different training objectives. The VAE compresses each trip and then tries to recreate it as closely as possible. Because it is scored on average accuracy, it favours typical, middle-of-the-road trips over rarer ones at the edges, where mistakes cost more. This pulls its output towards the centre of the distribution, as evidenced by the spread ratio of 0.804 and visual compression in spatial plots (Figure 5, Appendix C). By contrast, the WGAN-GP is trained against a critic that scores how closely the synthetic distribution matches the real one; this objective explicitly penalises mode collapse, rewarding the model for covering the breadth of real data. As a result its spatial distribution stays closer to the real dispersion, evidenced by the spread ratio 0.908.

Both generative models evidenced their ability to generalise to unseen data points, generating trips to and from location cells not seen in the original dataset. This is particularly evident by trips to/from locations outside of the Beijing boundary (observable in Figure 5), locations for which data was intentionally omitted from the original dataset during the pre-processing stage. This evidences the generative models' capability of interpolating from a continuous manifold, in contrast to bootstrap models method of sampling from real trips only.

However, while some generalisation is desirable, over generalising affects utility, evidenced by the lower OD-flow coverage scores - a trip from one specific origin cell

to another specific destination cell aligned with a real trip 42% of the time for the VAE model (in comparison to 67% for the benchmark). Whereas, WGAN-GP OD coverage was 37%, despite the lower spatial compression than the VAE. Therefore the generative models achieved as little as half the coverage of the benchmark evidencing their strong tendency to generalise to new location under the implemented representation in the continuous coordinate space.

All utility metrics had a relatively small standard deviation, showing robustness to initialisation. This is due to the large number of data points used for the statistical calculations, in contrast to the obtained privacy metrics which used few number of data points after the required user data splits, resulting in greater sensitivity to experimental noise.

Regarding privacy, the VAE and WGAN-GP achieve substantially better protection, with AUC scores close to 0.5 in comparison to the Bootstrap benchmark. The Bootstrap control was shown to be a positive control comparison, with the AUC of 0.75 showing strong success and advantage for the attacker. This gives credibility to the results of the MIA assessment, showing the attack is not broken. The Bootstrap AUC of 0.75 is lower than 1.0, showing that discretised OD trips are not always unique to individuals and that there is some ambiguity for the attacker.

However the demonstrated privacy for the generative models is at the cost of lower utility, as indicated in Figure 12. The x-axis shows the OD-flow Jensen-Shannon (JS) distance, where lower values indicate greater similarity to the real OD-flow distribution, while the y-axis shows the pooled AUC, where lower values indicate greater privacy and $AUC = 0.5$ indicates no membership leakage.

The membership inference results for RQ3 constitute an underpowered test, stratifying by activity level left as few as 28 users per group, producing confidence intervals too wide to support firm conclusions. The pooled AUC for the generative models shows a slight monotonic decline as activity increases, the opposite direction to the RQ3 hypothesis. However, the intervals are large and overlap the chance line ($AUC=0.5$) throughout, the apparent trend is most likely from sampling noise rather

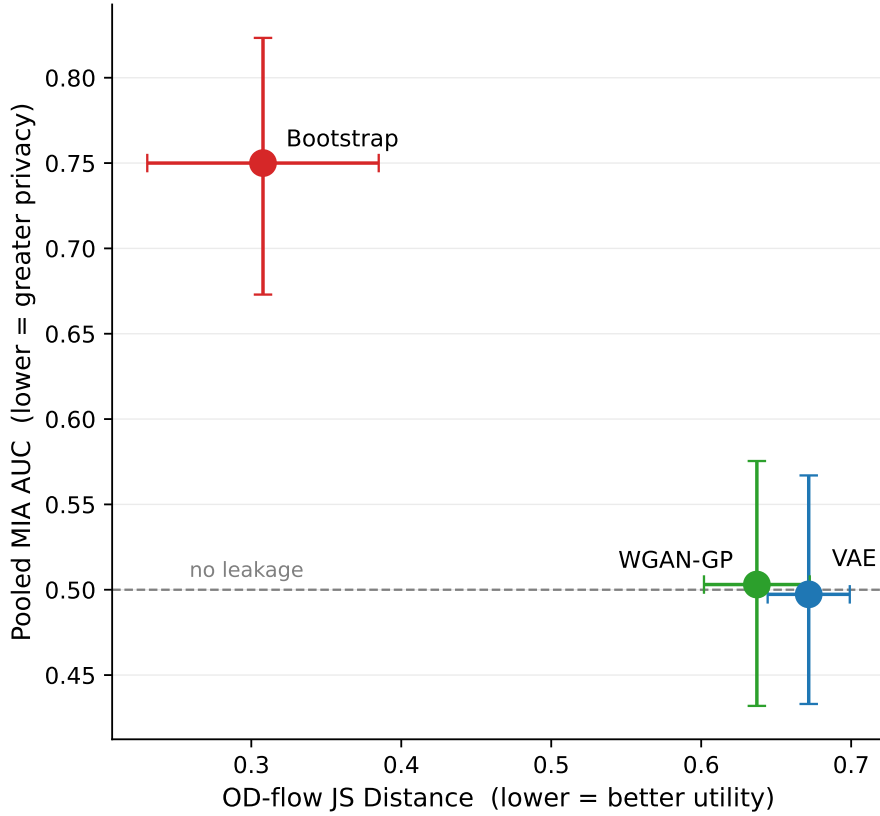


Figure 12: Privacy-utility trade-off between OD-flow fidelity and membership leakage. Error bars denote ± 1 standard deviation in OD-flow JS distance across seed runs and 95% cluster-bootstrap confidence intervals for the pooled MIA AUC.

than a real effect. The confidence interval of activity groups overlap substantially, no relationship between membership-inference accuracy and user activity can be established.

In contrast, the direct record reproduction shows a clearer story. The near-copy rate (the proportion of synthetic trips that closely match a trip in the real data) behaves consistently across all models (Figure 11). Notably, all models show the same monotonic pattern: the near-copy rate rises as user activity increases. Because this trend holds across both the baseline and the generative models, it appears to be a property of the data rather than of any single architecture. Users who contribute more trips are more likely to have at least one closely reproduced, indicating that higher-activity users carry a greater record-level reproduction risk. For the generative models this risk remains negligible in absolute terms (near-copy rates around

1%), but the consistent gradient suggests it would grow with activity were the models to fit the data more closely.

It is worth reconciling this gradient with the null membership-inference result, which might otherwise appear to contradict it. The two metrics measure different things: membership inference tests whether an attacker can infer a user’s presence and the deep models learned too little for any such signal to exist. By contrast, the near-copy rate measures incidental proximity between synthetic and real records, which rises mechanically with the number of trips a user contributes: the more trips a user has, the greater the chance that at least one synthetic trip lands near one of them, whether or not the model memorised anything. A rising reproduction risk alongside no membership leakage can therefore be consistent and not contradictory.

5.2 Comparison to Prior Work

The central finding — that synthetic data does not automatically yield a better privacy–utility trade-off — aligns with Stadler, Oprisanu, and Troncoso (2022), who argue that the privacy gains of synthetic data are unpredictable relative to traditional anonymisation, making both the privacy benefit and the utility cost difficult to guarantee in advance. The present results are consistent with this: privacy emerged not as a reliable property of the method but as a by-product of under-fitting.

The low-leakage outcome also echoes Yuan et al. (2026), whose WorldMove models likewise report membership inference below chance-adjacent levels (< 0.6). The agreement is notable precisely because the cause differs. WorldMove achieves low leakage through large-scale aggregation across thousands of cities, whereas the low leakage here arises from a model that has not fit its (small) training data closely enough to memorise it. The same headline number is reached by opposite routes — strong aggregation in one case, weak fitting in the other — which cautions against reading a low MIA score as evidence of a well-designed privacy mechanism.

The activity-level results relate more tentatively to prior observations of uneven per-user risk. Stadler, Oprisanu, and Troncoso (2022) find that outliers are system-

atically under-protected, and Webster et al. (2021) show that identities contributing more samples are more readily re-identified. The near-copy gradient reported here — record-reproduction risk rising with user activity — is directionally consistent with both. The membership inference results, however, cannot corroborate them: with no measurable memorisation in the deep models and an under-powered stratified test, there is simply no leakage signal to resolve. This study therefore extends the observation of uneven risk to the record-reproduction channel rather than confirming it through membership inference.

Finally, the work sits apart from the differentially private, tabular approaches typified by Abay et al. (2019). Where that line of work provides formal guarantees on tabular data, this study offers a complementary counterpoint: no formal DP, but a user-level membership inference evaluation on spatiotemporal origin–destination data, with a copying positive control establishing that the attack can detect leakage when it exists.

5.3 Limitations

The Dataset - The majority of users in this dataset have a small number of trips, this could be considered unrealistic for large-scale dataset publication. Whereas, more trips per user would provide a greater assessment of privacy in which generation models have a more opportunity to memorise distinct user behaviour patterns, and therefore pose a higher risk of privacy leak. However, the limited access to large open datasets for mobility only reinforces the motivation of this research, helping to enable more publications of data.

The dataset size also shaped the spatial representation: Origins and destinations were represented as continuous centroid coordinates, a choice that preserved a usable spatial metric for the generative models. Continuous coordinates impose geographic smoothness as a proxy for behavioural similarity, treating spatially adjacent cells as interchangeable and permitting the models to interpolate into unobserved regions, as observed with the generated models. The use of learned embeddings instead

would help respect the discrete structure of real-world travel behaviours, however embeddings require substantially more data than was available here, but this type of representation would be a natural direction for future work.

The implemented membership inference attack relies heavily on spatial features, supported by other features such as trip duration and trip count. The focus on spatial features reflects the locational nature of mobility-based re-identification, but means the attack does not exploit temporal or sequential structure to the same degree — for instance the timing or ordering of a user’s trips. A richer feature set incorporating such signals could recover additional leakage, extending the attack in this direction is left to future work.

Statistical Confidence - Furthermore, the statistical strength of the MIA was lower than the other assessments performed due to the small number of users evaluated after conducting various train-test and IN/OUT data splits. This is particularly the case for RQ3, where stratification by activity level for the MIA assessment resulted in as little as 28 users per group per model, causing wide AUC confidence intervals despite pooling across seed runs.

This restriction even applies to some metrics in the unstratified RQ2 assessment — with roughly 85 users in the evaluation pool, a 1% false-positive rate (as reported with TPR @1% FPR) corresponds to a lower resolution than a single user, placing the operating point below the granularity the data can support. The true-positive rate at this threshold therefore changes in discrete jumps rather than smoothly, and cannot be estimated at the precision its nominal threshold implies.

The Models assessed occupy extremes of the utility range, where the Bootstrap baseline reproduces the real dataset extremely closely, while the under fit generative models alter the data significantly; no model sits in the intermediate regime. As a result, the relationship between utility and privacy can only be established at these two extremes but not traced in-between. While evidence of protected privacy is a requirement, an under-fit model with low utility is not a deployable real-world mechanism. Future work would include further model training and hyper-parameter

searches, as well as additional model architectures to better explore model performance of generative architectures and assess the utility-privacy trade-off across a wider range of models.

Alternative Evaluation Methods - Finally, membership inference was used as the primary assessment of privacy, but it captures only one type of disclosure risk. A more comprehensive evaluation would incorporate complementary attacks, such as attribute inference, in which an adversary with partial knowledge of a record attempts to infer its remaining sensitive attributes or reconstruction attacks, as described by Salem et al. (2023). Establishing privacy across this broader range of threat models is a necessary step before synthetic mobility data of this kind could be released with confidence.

Furthermore, future work could layer the generator with the explicit statistical disclosure control methods as discussed in section 2.4 — post-hoc filtering of near-copy synthetic records, output-level noise injection, or most rigorously, differentially private training — to convert privacy from an incidental by-product of underfitting into an engineered, verifiable property of the release pipeline.

5.4 Conclusions

This study asked whether deep generative models can produce origin–destination mobility data that is both useful and private. The motivating hope was that the two could be decoupled, achieving both desirable traits together without compromise. Instead, the implemented deep models achieved strong privacy at a substantial cost to utility, indicating that the privacy–utility relationship can be understood as a trade-off mediated by model fit rather than independent attributes that synthetic data grants for free.

On utility (RQ1), the generative models reproduced temporal structure closely, with low Jensen–Shannon distances for the hour-of-day and day-of-week distributions (0.12–0.15), and realistic trip start times and durations. Spatial structure was reproduced less faithfully: origin and destination cell distributions diverged sharply from the real data (JS distance > 0.8), and the downstream destination-prediction task showed a heavy loss of utility (TSTR ratio 0.25–0.27, against 0.99 for the bootstrap). The two models differed in how they handled spatial spread: the WGAN-GP preserved the real dispersion slightly better, whereas the VAE compressed trips toward the urban centre, as visible in the spatial plots and reflected in the lower spread ratio. Both models generalised readily to unobserved locations, a natural direction for future work should be adopting a learned embedding as an alternative data representation for the models, better suiting the discrete nature of real-world travel behaviours.

On privacy (RQ2), a user-level membership inference attack was evaluated against a bootstrap positive control. The control confirmed a functioning attack, exhibiting clear leakage (AUC 0.75); its ceiling below 1.0 reflects the ambiguity introduced when discretised OD trips are shared across individuals rather than unique to them. Against this reference, neither generative model showed measurable leakage (AUC ≈ 0.50) or direct record reproduction (near-copy rate below 1%).

On differential risk by activity level (RQ3), the membership inference results were inconclusive: stratifying by activity left as few as 28 users per group, producing con-

fidence intervals too wide to support any claim, a larger dataset would be needed to give the test sufficient statistical power. However, the near-copy rate rose monotonically with user activity, consistent across all models including the bootstrap. As a property of the data rather than of any single architecture, this indicates that higher-activity users carry a greater record-reproduction risk in principle.

Taken together, the implemented deep models behaved as under-fit generators: private but with limited utility. An under-fit model is not a deployable privacy mechanism, the absence of measurable leakage does not by itself mean data is safe to release. Because the bootstrap and the generative models occupy opposite extremes of the utility range, the trade-off is established only at its endpoints; training intermediate models to trace the region between them is an important next step. Finally, the study underlined how a small dataset constrains the choices available for data representation, model design, and ultimately utility–privacy performance — the scarcity of large and openly accessible mobility datasets only reinforces the importance of this research area.

The evaluation pipeline built here — near-copy rate, user-level MIA, and activity-stratified analysis — provides the tooling needed to verify privacy in future models that pursue utility more aggressively, whether through improved architectures or through explicit statistical disclosure control mechanisms such as differential privacy. This work lays the methodological groundwork for synthetic mobility data that is both useful and demonstrably private.

AI Usage Statement

AI tools, specifically ChatGPT 5.4 and Claude Opus 4.8, were used for the drafting of experimental code for the project, as well as for the identification of errors and bugs. Architectural and experimental design choices were taken by the author. All code was checked and reviewed by the author, who takes full responsibility for the implementation.

In addition, the AI tools were used for some refinement of language used in this report, as well as helping identify relevant academic material for the project. All papers referenced in the report have been read and analysed by the author.

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Appendix A

Preprocessing and Trip Segmentation

All data was separated by user ID, an anonymous identifier distinguishing each volunteer’s GPS-logged data. Data was further segmented by session ID, representing each period from the user switching on their GPS logger until switching it off.

Once partitioned by session, a simple logic was used to identify static periods with minimal movement; these ‘stay points’ were then used to segment session trajectories into multiple trips, each stay point representing either the origin or destination of an origin–destination (OD) trip. If GPS movement remained within 100 m of a rolling-average location for a 20 min period, a stay point was identified and the session trajectory segmented. The method also accounted for GPS signal accuracy, which can vary significantly in dense urban environments. This procedure is semantically similar to previous studies on the dataset, in which a stay point is identified when movement stays within a given distance threshold for a given time — for example 200 m within 30 min, as used by Li et al. (2008). The use of a rolling-average makes the process more resistant to GPS noise, therefore the parameters of 100 m for 20 min were chosen (all points within 100 m of the rolling average)

The algorithm was implemented conservatively: trajectories were only segmented when there was sufficient data to confirm that GPS movement remained within the threshold (at least 5 GPS data points needed), since GPS data may be sparse or absent when users travel underground or by metro. As a backstop, very long gaps

between consecutive GPS points — greater than 30 min — were also used to segment trips. Algorithm 1 describes the segmentation algorithm.

Algorithm 1 Trip segmentation via stay-point detection

```

1: for each user ID do
2:   for each session ID do
3:     sort records by timestamp
4:     for each gap between consecutive timestamps > 30 min do
5:       segment session into separate trips
6:     end for
7:     for each rolling window of  $\geq 5$  GPS records do
8:       compute rolling-average centroid
9:       compute haversine distance of each point to the centroid
10:      if all points stay within 100 m for 20 min then
11:        segment session into separate trips
12:      end if
13:    end for
14:  end for
15: end for

```

To further reduce GPS location noise and to produce more clearly defined OD trips, a spatial aggregation step was applied: the start and end coordinates of each trip were snapped to the nearest grid centroid. Discretising locations in this way suppresses residual GPS noise, with each trip origin and destination represented by the centroid of its grid cell. The H3 hierarchical hexagonal grid system was used, which can be implemented across different resolutions (*H3: Uber’s Hexagonal Hierarchical Spatial Index* 2018). Hexagonal cells were preferred to a square grid for their directionally uniform centre-to-centre distance. Resolution-6 is used throughout, including in the spatial plots such as Figure 5 and Figure 15. Cells at this resolution cover approximately 36 km² on average (Uber Technologies n.d.), chosen as fine enough to capture individual travel behaviour yet coarse enough that locations recur, giving the models learnable shared flows rather than a sparse field of unique endpoints.

Trip start time was defined as the last GPS timestamp recorded within the origin cell, and trip end time as the first timestamp recorded within the destination cell. Because users often linger near their origin and destination, taking the first and last points of the trajectory would absorb this dwell time into the trip duration. Anchoring the boundaries to cell transitions instead excludes time spent stationary

within the origin and destination cells, so the recorded duration more closely reflects the movement between them. Trips inferred as starting and ending in the same grid cell were removed from the dataset.

Although the original dataset is predominantly based in Beijing, some trajectories relate to travel outside the city. All trips to or from locations outside Beijing were therefore removed, focusing the dataset on a single city and restricting the coordinate variation of trips. The standard GADM level-1 boundary was used as the definition of Beijing (GADM n.d.).

The resulting processed dataset is a list of OD trips, which include a start and end coordinates (representative of the grid cell centroid), as well as trip start and end timestamps, along with the associated user id for the trip.

Appendix B

Hyperparameter Selection

Hyperparameters were not selected through an exhaustive grid search but through a manual exploration, guided by the utility metrics evaluated on held-out data. The two generative models had some fixed shared training settings such as the optimiser and batch size. The training loss curves shown below were used only to confirm stable convergence and are not a measure of synthetic-data quality.

After tuning, the epoch count was reduced where performance ceased to improve in order to avoid unnecessary training — a particular consideration given that multiple shadow generators were also trained per experiment. Limited computation and time meant practical compromises were made to keep each experiment within a feasible budget.

B.1 Variational Autoencoder

Exploration of the VAE centred on the two settings most directly responsible for its tendency to over-smooth the spatial distribution: the latent dimension and the KL weight β . Reducing the latent dimension was found to lessen this over-smoothing: on this small dataset a larger latent space was only partially used, leaving the sampled distribution contracted, whereas a smaller latent was more fully utilised and better preserved spatial spread. A latent dimension of 8 was selected; although

the reduced capacity came at some cost to overall distributional fidelity, it most effectively mitigated the spatial contraction that was the VAE’s primary weakness.

The KL weight β was raised to 0.1, rather than the initial 0.001. At very low β the latent posterior diverged from the prior, so that sampling from the prior at generation time produced poorly-formed, contracted outputs; increasing β improved the match between the aggregate posterior and the prior and reduced this contraction. A linear KL annealing schedule was added, ramping β from zero to its final value over the first 40% of training epochs; this improved stability by allowing the model to establish useful reconstructions before the latent regularisation was applied at full strength. Figure 13 show the training curve losses for a single seed run (seed 111). The KL term remains non-negligible as β increases, indicating the latent space is used rather than collapsed. Table 3 summarises the final configuration parameters selected.

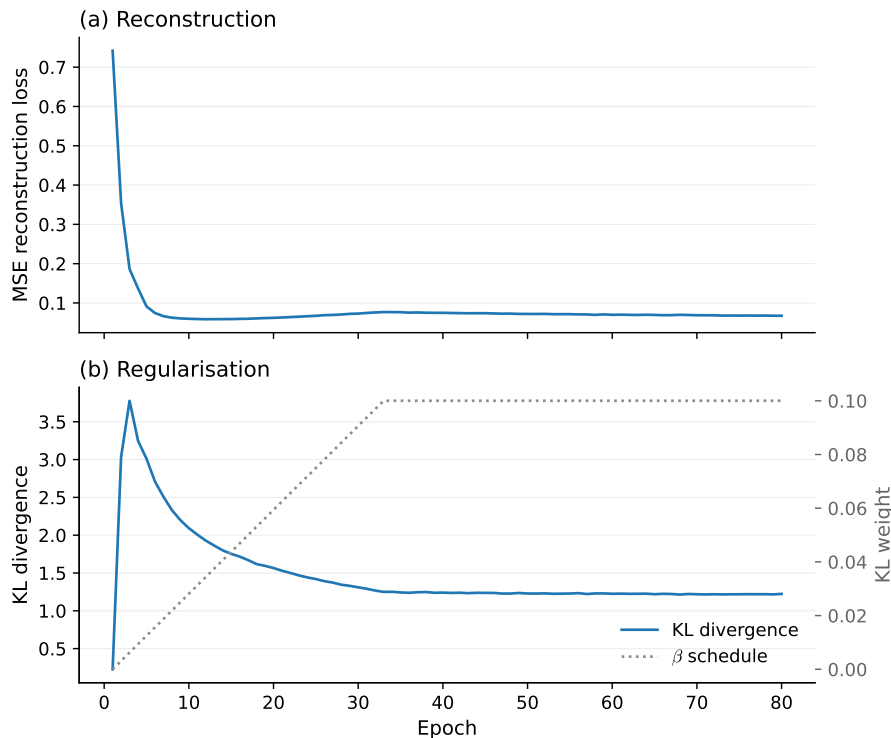


Figure 13: VAE training curves (seed 111). **(a)** Mean-squared-error reconstruction loss on the training and validation splits. **(b)** Kullback–Leibler divergence, with the linear β annealing schedule overlaid (grey).

B.2 Wasserstein GAN with Gradient Penalty

For the WGAN-GP, the number of critic steps and the hidden dimension were the main settings explored. Reducing the critic steps from the standard five to three, and briefly to two, shortened training time with little observed loss in utility; three was selected as a practical compromise, which was valuable given the repeated training of shadow generators. The hidden dimension was varied, with the larger width (128) selected. Unlike the VAE, the WGAN-GP continued to show gradual improvement across the training budget rather than plateauing, and was therefore trained for a longer 100 epochs. Further epochs would have been added but experiment training time had to be limited for practical reasons.

Figure 14 shows training loss for both generator and critic for a single seed run (seed 111). The critic loss approximates the negative Wasserstein distance and is expected to take negative values; both traces stabilise within the training budget. These curves characterise training dynamics only and do not measure synthetic-data quality. Table 3 summarises the final configuration parameters selected.

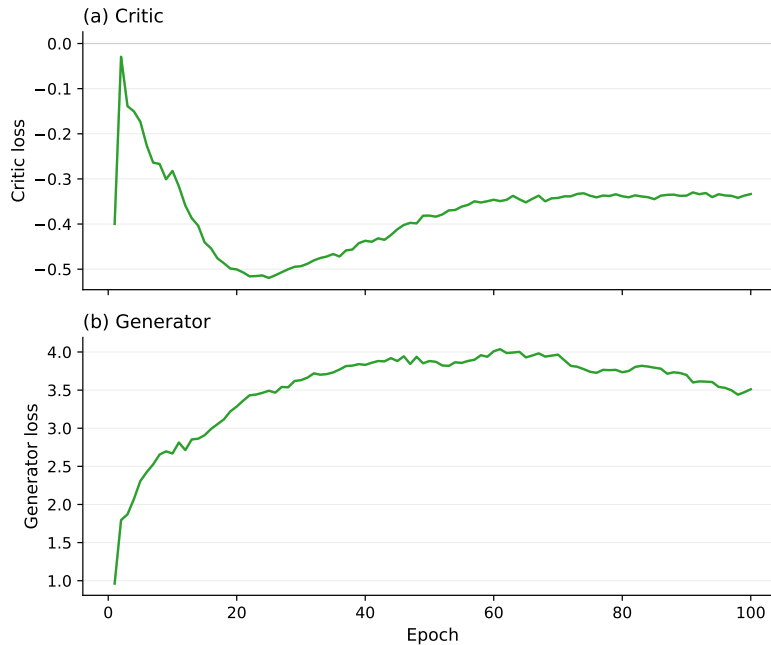


Figure 14: WGAN-GP critic and generator loss by epoch (seed 111).

Appendix C

Spatial OD Plots

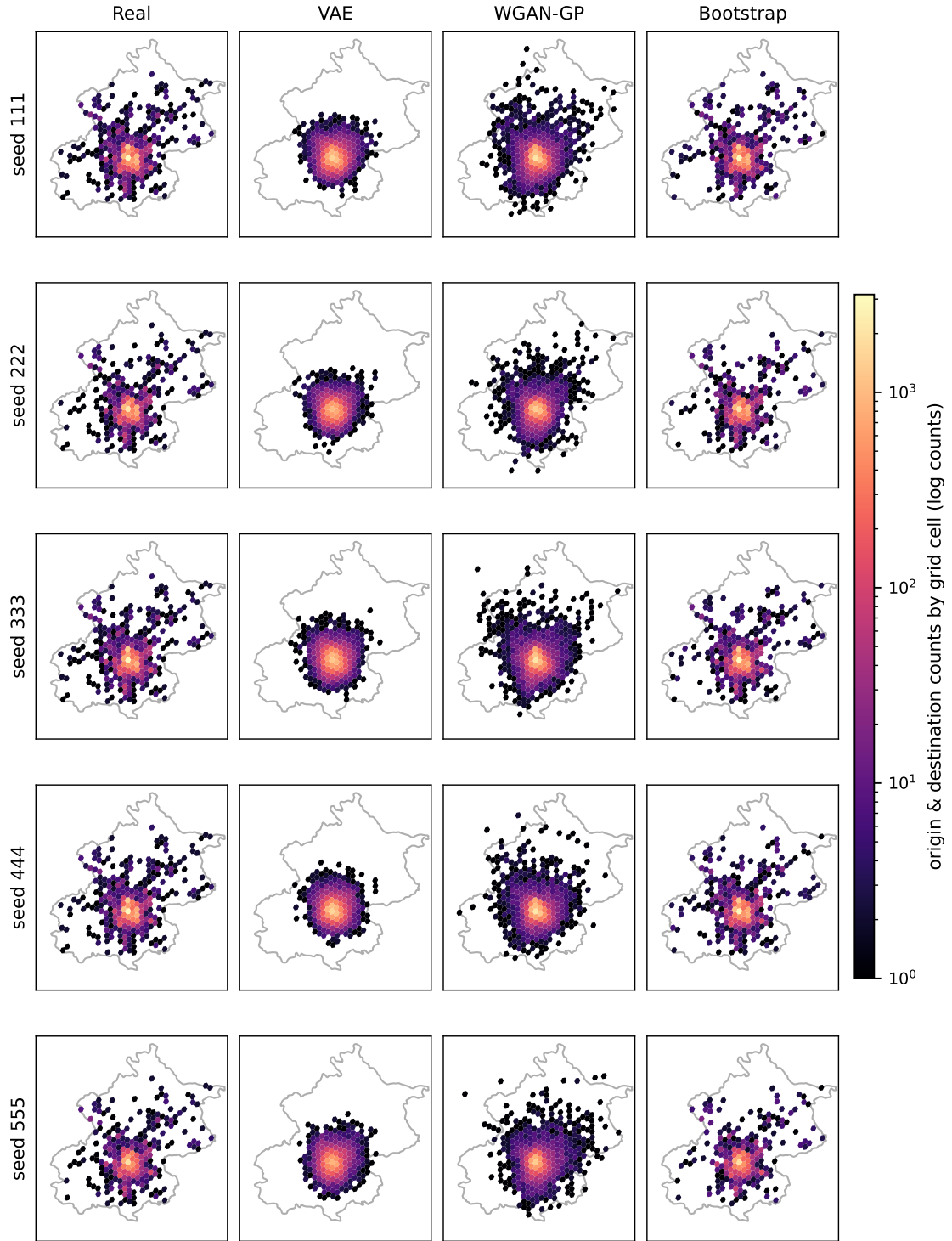


Figure 15: Spatial distribution of trip endpoints across all five seed runs, for the real training-split data and each generative model. Each panel maps the log count of combined origin and destination points falling within H3 resolution-6 hexagonal grid cells, on a shared colour scale across all panels. The grey outline is the GADM 4.1 administrative boundary of Beijing municipality.